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Joint inference of mutational signatures from indels and single-nucleotide substitutions reveals prognostic impact of DNA repair deficiencies

Patricia Ferrer-Torres¹, Iván Galván-Femenía¹ and Fran Supek^{1,2,3*}

Abstract

Background Mutational signatures are increasingly used to understand the mechanisms causing cancer. However, their important applications in predicting prognosis and stratifying patients for therapy are hampered by inaccurate inference of the various featureless, dense trinucleotide mutational spectra, which are often confounded with one another. One of them is the homologous recombination deficiency (HRd)-associated signature SBS3, relevant because of its association with prognosis in ovarian and breast cancer and because of its potential as a biomarker for synthetic lethality therapies.

Methods Here, we highlight strong benefits of a multimodal approach for mutational signature extraction, applied on top of standard bioinformatic pipelines. By jointly operating on single-base substitution (SBS) and indel (ID) spectra, this method enables accurate identification of various DNA repair deficiency signatures and patient survival prediction.

Results Across four different cohorts of whole-genome sequenced high-grade serous ovarian cancers (HGSOC), the multimodal SBS + ID approach correctly distinguished the commonly confused signatures SBS3, SBS5, SBS8, SBS39, and SBS40. Importantly, we robustly identified two different multimodal SBS3 signatures, m-SBS3a and m-SBS3b, with distinct patterns in the indel spectrum. Multimodal SBS3b signature was strongly predictive of longer survival in ovarian cancer patients, replicating across four cohorts, with effect sizes greatly exceeding other genetic markers. Our m-SBS3 also predicted survival in platinum-treated patients with various cancer types, and moreover, the SBS + ID joint inference was successfully applied to mismatch repair-deficient colorectal cancer and immunotherapy response, supporting a general utility of the multimodal mutational signatures approach.

Conclusions Overall, combining SBS and ID mutations improves detection of HR deficiency-associated signatures and reveals distinct SBS3 subtypes with prognostic value. This multimodal approach outperforms existing markers and is readily applicable to therapy stratification.

Keywords Mutational signatures, Somatic mutations, Indels, Ovarian cancer, Homologous recombination deficiency, Mismatch repair deficiency, Prognostic marker

*Correspondence:

Fran Supek

fran.supek@bric.ku.dk

Full list of author information is available at the end of the article



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Background

As somatic cells progress towards malignancy, they are exposed to different mutational processes that leave overlapping, multi-scale patterns of somatic mutations across the genome. Mutational processes can result from a combination of DNA damage, DNA repair, and DNA replication mechanisms, both naturally occurring and pathological [1, 2]. These factors can have an endogenous or an exogenous origin, caused by germline variations, driver mutations, epigenetic modifications, and environmental or therapeutic exposures to mutagens.

After sequencing, the thousands of somatic mutations present in a patient's tumor can be classified by the trinucleotide or pentanucleotide context, deriving a mutational spectrum. Such spectra can be deconvolved into mutational signatures, which aim to identify biological mechanisms that generated the mutations in a given tumor genome, and quantify their relative activities.

Mutational signatures provide valuable insights into the processes driving tumorigenesis [2–4], which can carry clinical relevance, acting as biomarkers for therapy sensitivity and prognosis independently of known clinical and molecular biomarkers [3]. Estimating the mechanisms that generate the signatures can be challenging due to various technical factors in identifying mutations and also limitations in the algorithms for signature inference. While signatures can be decoded and mapped to causal mechanisms using controlled mutagenesis experiments [5–11], the signatures identified from large-scale observational analyses of genomes are prone to errors, and usually do not have clear causes.

Our study is focused on improving the identification of mutational signatures occurring in cancer genomes, and assessing their utility in predicting response to chemotherapies and patient survival. This is motivated by studies showing that mutational signatures are generally robust predictors of drug response in cancer cell line screens [12], and that mutational signatures of homologous recombination (HR) deficiency, APOBEC hyperactivity, and deficient DNA mismatch repair (MMR) imply the existence of tumor vulnerabilities that may be exploited therapeutically [12–16].

We focus on the most common and aggressive type of ovarian cancer, high-grade serous ovarian carcinoma (HGSOC). This malignancy regularly develops resistance to chemotherapy, resulting in a 5-year overall survival rate of less than 40 % [17, 18]. HR deficiency (HRd) affects many HGSOC patients, by some estimates ~50 % [19, 20]. One common cause are germline pathogenic variants in *BRCA1*, *BRCA2*, *PALB2*, *RAD51C*, *RAD51D* and potentially other related genes [21–25]. Other causes

including somatic copy number alterations (CNA) and epigenetic silencing of these genes, or potentially mutations in other genes can also disrupt HR. Many of these mutations have unclear impact, making it difficult to ascertain HRd just based on the presence of known pathogenic variants in a tumor genome. HRd can however also be ascertained using genome-wide patterns of somatic mutations. DNA repair deficiencies leave different types of mutational footprints in the genome, in the case of HRd including most prominently the mutational signature of single-nucleotide variants (SNVs) SBS3 [1, 26–28]. The trinucleotide spectrum of SBS3 (previously called Signature 3) is described as “flat” or “featureless,” meaning it has contributions from various mutation types and trinucleotide contexts. This property increases the risk of confounding with other signatures that have similarly “featureless” trinucleotide spectra such as the very widespread signatures SBS5 and SBS40, as well as the SBS8 associated with nucleotide excision repair (NER) pathway deficiencies [2].

While the exact origin of the SBS3 mutations is still to be ascertained, they are likely generated by HRd only indirectly, resulting from a downstream process; translesion DNA polymerases were proposed to directly underlie SBS3 [29, 30]. Thus, SBS3 can arise in the genome even in the absence of HRd, as observed in stomach cancers without obvious *BRCA1* and *BRCA2* pathogenic variants [31]. In the same vein, in breast and ovarian cancer, various SBS3 cases cannot be explained by germline and somatic variants and epigenetic silencing [32], suggesting that some of these tumors might not be fully HRd as one explanation. Another explanation, not mutually exclusive, involves unrecognized pathogenic variants or epigenetic changes in HR-adjacent genes [25]. Overall, refining the process of extracting SBS3 may lead to a more precise assessment of HR functional status in tumor samples.

An opportunity for this arises because, in addition to the SBS3 spectrum of point mutations, HRd generates footprints of other types of genetic changes. On the one hand, this includes different large-scale rearrangements, resulting in copy number deletions and amplifications of certain size ranges. On the other hand, HRd also results in indels, characteristically small deletions of >5 bp, with short microhomology at flanking regions, resulting from increased use of alternative double-strand break pathways when HR is deficient. This type of indel is captured in the ID mutational signatures ID6 and ID8 from the COSMIC catalog, which were found associated with alterations in HRd genes across cancer genomes [1].

For the task of identification of HR deficient patients, rather than necessarily using SBS signatures on their own [28, 33], previous methods generated machine learning classifiers (HRDetect and CHORD methods [23, 34])

where SBS signatures were combined with rearrangement patterns and indels, to predict mutational status of *BRCA1* and *BRCA2* genes as a ground-truth. In principle, the identification of HRd tumors should help for assigning patients to therapies, as HRd cells are sensitized to poly(ADP-ribose) polymerase inhibitors (PARPi) [35] and also respond better to standard platinum chemotherapy [36]. However, such HRd classifiers are specific to prediction of *BRCA1/BRCA2* functional status, and do not generalize to other mutational processes. Moreover, their utility for predicting patient response robustly, replicating across various cohorts of HGSOC and other cancer types, remains to be shown.

Here, we propose a generally applicable, unsupervised method for mutational signature extraction, where “multimodal” signatures are jointly inferred from SBS and ID spectra. This is based on a simple modification of existing non-negative matrix factorization (NMF)-based tools and applies to de novo extraction as well as signature refitting. The multimodal approach improves detection of the featureless signature SBS3 and demonstrates it should be separated into two sub-signatures, SBS3a and SBS3b. The multimodal SBS3b signature highly accurately predicts survival in ovarian cancer patients, replicating across 4 HGSOC cohorts, and also has prognostic relevance to platinum-treated pancreatic, prostate, and breast cancers. Furthermore, since the multimodal approach does not require ground-truth labels, it can be applied to any mutagenic process. We demonstrate how its application to MMR deficiency (MMRd) signatures improves prediction of survival and immunotherapy response in colorectal cancer.

Methods

Datasets

We used somatic mutation calls (SNV and ID) from four independent cohorts of HGSOC WGS samples: DECIDER[37] ($n = 100$), Pan-Cancer Analysis of Whole Genomes (PCAWG) [38] (OV-AU and OV-US) ($n = 110$), Hartwig Medical Foundation [39] (Hartwig) ($n = 124$), and OVCARE [40] ($n = 52$). The DECIDER cohort is composed fully of high-grade serous ovarian cancer (HGSOC) samples. For Hartwig and OVCARE, the HGSOC samples were selected. In the PCAWG cohort database of ovarian cancers, subtypes were not listed to our knowledge; however, we found information that samples from PCAWG are classified as ovarian serous adenocarcinoma [38, 41]. Although low-grade or high-grade was unspecified, 90–95 % [42] of ovarian serous adenocarcinomas are high-grade and this is the more aggressive subtype; we thus infer that also for PCAWG, most samples will most likely be from the high-grade variety. Moreover, other publications have considered

PCAWG ovarian samples as high-grade serous ovarian cancer [43]. Note that the previously reported sample sizes are for the selected subset of high-grade serous cancer samples. DECIDER, PCAWG, and OVCARE cohorts correspond to primary ovarian tumor samples, while Hartwig is composed of metastatic samples.

For prostate, TN-breast, and pancreatic cancer, WGS samples from Hartwig (metastatic tumor samples) were used ($n = 165$, $n = 78$, and $n = 384$, respectively) [39]. In the case of breast cancer samples, only those labeled as triple-negative (TN) breast cancer were selected for the analyses, because of the higher relevance of HR deficiencies to TN breast cancers. For colorectal cancers, a total of 1800 WGS samples were used, 1063 from the Uppsala University large-scale WGS Swedish cohort [44] (primary tumors), and 737 from Hartwig [39] (metastatic tumors).

Extraction of mutational signatures: SigProfiler

The de novo extraction of mutational signatures is an unsupervised machine learning method to extract mutational signatures that are present in a catalog of mutations obtained from a set of samples. SigProfiler tool [45, 46] was used to perform de novo signature extraction. The first step in the extraction is to run SigProfiler MatrixGenerator [45], which processes input *vcf* files containing the somatic mutations present in a cohort of cancer genomes. The result is the M matrix, which is composed of the counts of different mutation types. For standard signatures, the SBS96 mutation count matrix was used. This matrix is composed of 96 SBS channels or mutation types, depending on the trinucleotide context of the SNV mutation. For multimodal signatures, we joined the SBS96 matrix to the ID83 matrix. The latter is composed of 83 ID mutation types, depending on the length, the type, and the context of the ID mutation. These two matrices were merged into a SBS + ID mutation count matrix, where the spectrum is composed of 179 channels (96 from SBSs and 83 from IDs mutation types).

These matrices were used as input for SigProfiler Extractor [46] which performs NMF to approximate the M matrix to the product of two matrices, the signature matrix S, and the exposure matrix E. Matrix S corresponds to the SBS and ID profiles of the signatures extracted from the input. In the case of the standard signatures, the dimensions of the S matrix are 96 by the number of signatures extracted (k). In the case of the multimodal SBS + ID signatures, the dimensions of the S matrix are 179 by k . The E matrix corresponds to the activities of those signatures in the tumor genome samples. The dimensions of the E matrix are k by n , where n is the number of samples. The minimum and maximum number of extracted signatures were set to 1 and 30.

Additionally, the number of iterations to be performed per extraction process was set to 100.

We used the number of signatures k suggested by SigProfiler based on mean sample cosine distance and average stability. We found that, generally, the number of multimodal signatures extracted was higher than the number of standard (SBS-only) signatures in the same cohort. This can be explained because adding indels can differentiate mutational processes, and the extracted multimodal signatures often comprised also indel mutations. In our pan-HGSOC analysis, $k = 14$ was suggested for both standard and multimodal methods. After looking at the SigProfiler selection plot for multimodal signatures, we selected $k = 20$ for multimodal signatures, which also had a high average stability (around 0.8) and a low mean sample cosine distance.

Next, we matched the de novo signatures to COSMIC version 3.3 reference signatures (<https://cancer.sanger.ac.uk/signatures/>). For the standard SBS signatures, SigProfiler Extractor automatically outputs the COSMIC reference signatures that best match the de novo extracted signatures. However, for the multimodal SBS + ID signatures, this cannot be done automatically, as no reference multimodal SBS + ID signatures exist. To do so, the profiles of the 179-channel signatures were divided into the SBS part (96 channels) and the ID part (83 channels). Then, the *decompose_fit* tool from SigProfilerAssignment [47] was used to find the corresponding COSMIC signatures for the SBS and the ID parts of the signatures separately.

We considered a mutational signature to be identified when SigProfiler individually assigned it to one of the extracted patterns and it represented the major percentage of the signature. For example, if signature SBS96A was assigned to SBS8 (85 %) + SBS3 (15 %), we would label this signature as SBS8. If a signature was assigned to two signatures in a similar percentage, we would consider these signatures mixed. For example, if signature SBS96B was assigned to SBS8 (55 %) + SBS3 (45 %), we would consider it a mixture of SBS8 and SBS3. Generally, we stated that a signature was mixed when the major signature represented less than 80 % of the profile.

We extracted both standard and multimodal signatures from the four HGSOC cohorts together, referred to as pan-HGSOC. Moreover, we also extracted mutational signatures in each of the individual cohorts: DECIDER, Hartwig, PCAWG, and OVCARE to compare how the number of samples impacts on the extraction of our signatures of interest. For non-ovarian cancer types, we extracted signatures in the individual Hartwig cohorts of TN breast cancer, prostate, and pancreas. Finally, mutational signatures were extracted from two combined

datasets of colorectal cancer: Hartwig and Uppsala University cohorts.

Survival analysis

For the survival analyses, we transformed the signature exposures from a continuous to a binary categorical variable: signature-positive and signature-negative. This was determined based on whether a sample's signature exposure was higher or lower than a certain threshold. The thresholds were selected by calculating various percentiles of the activities of the signatures, where thresholds ranging from 20 to 80th percentile, were tested for association with overall survival. The percentile threshold with the strongest Z-score or difference in median survival was selected.

To explore the ability of the extracted signatures to predict the overall survival of ovarian cancer patients, we performed two types of survival analysis. First, Kaplan–Meier curves were plotted for the signature-negative and signature-positive patient subgroups. To do so, the *survfit* and *ggsurvplot* functions from the *survival* and the *survminer* R packages were used. We used the *surv_median* function to obtain the median survival for each of the groups, and the *surv_pvalue* function that performs a log-rank test to obtain the p -value. For the analysis of the pan-HGSOC signatures, we show the results where a 30th percentile is used for sample classification (Fig. 1e). We found the 30th percentile to strike a good balance between statistical significance and median survival difference. However, we also show the results for different thresholds in Additional File 1: Fig. S6.

The second type of survival analysis we performed was Cox proportional hazards regression analysis, in order to adjust the results by confounding factors. We refer to the analysis as univariate Cox when the regression included only one signature, as well as age and stage (and cohort when the signature exposures were obtained from more than one cohort jointly). For multivariate Cox, the exposures of the rest of the signatures extracted were also included in the regression, in order to adjust the results for the effect of other mutational signatures. We performed Cox regression using the *coxph* function from the *survival* R package, using the formula:

$$\text{Surv}(\text{survival}_i, \text{time}, \text{status}) \sim \text{presence}_{\text{sign}_1} + \text{presence}_{\text{sign}_2} + (\dots) + \text{presence}_{\text{sign}_k} + \text{age} + \text{stage}$$

In this case, for testing different thresholds, the percentile used to classify the signatures was fixed (results in heatmap plots are shown for 30 percentile), except for our signature of interest, whose percentile used for classification changed (this is represented in the x axis of the heatmap plots). We extracted Z-scores and adjusted p -values for the signatures from the *coxph* function to

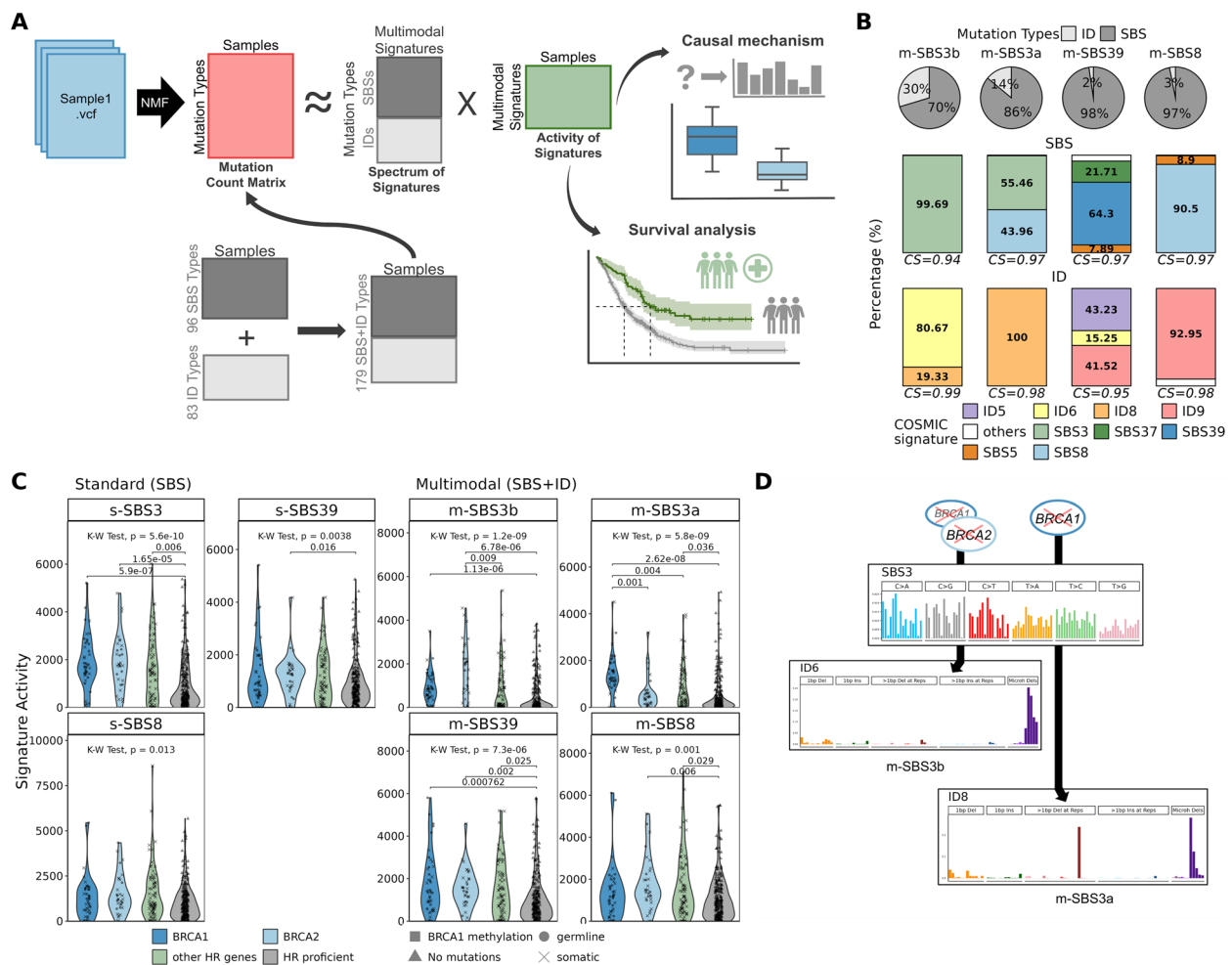


Fig. 1 Multimodal mutational signatures combining SBS and ID spectra more accurately distinguish HR mechanisms. **A** Multimodal signature extraction process consists in concatenating matrices for SBSf and ID mutation counts before performing NMF. Multimodal signatures are valuable predictors of survival and help clarify causal mechanisms. **B** SBS versus ID composition of 4 multimodal signatures extracted (above) and proportion of known COSMIC SBS (middle) and ID (bottom) signatures therein. **C** Association of standard and multimodal mutational signature activities with *BRCA1*, *BRCA2* and other HR genes (see gene list in Methods) damaging germline and/or somatic variants (Kruskal–Wallis test followed by a Mann–Whitney/Wilcoxon rank sum test). **D** Schematic diagram showing the composition and etiology of m-SBS3a and m-SBS3b

study the results. We used the function *cox2.display()* from the *jstable* R package to extract the hazard ratios (HR) and their 95 % confidence intervals (CI).

In order to check possible multicollinearity problems in our Cox regression models, we performed a variance inflation factor (VIF) analysis, including multimodal mutational signatures, age, stage, and cohort variables. All variables, except for cohort, showed VIF values between 1 and 3, indicating moderate multicollinearity. The highest VIF value (5.64) was observed for the PCAWG cohort, but overall, collinearity remained within acceptable limits.

We also calculated the concordance or C-index for each Cox regression. The concordance index is a helpful

metric, related to an AUC score, which measures the predictive performance of the model, specifically how well it predicts the correct ordering of survival times. Therefore, it allows us to truly compare the predictive ability of our mutational signatures.

Association between mutational signatures and duplications/deletion structural variants

Duplications and deletion structural variants (SVs) (> 50 bp) were extracted from the LINX calls for PCAWG and Hartwig patients and from ASCAT for DECIDER patients. For duplications, the “DUP” flag was used to obtain variants from Linx calls, whereas for Ascat, we selected variants with allele counts of nMajor > 1. For

deletions, homozygous states were identified using the JunctionCopyNumber < 0.5 filter in Linx and the nMinor == 0 & nMajor == 0 filter in Ascat.

We tested for association between the occurrence of duplications and deletions, and activity of specific signatures: m-SBS3a, m-SBS3b, s-ID6, and s-ID8. We grouped the duplications and deletions across different segment lengths, following the classification used in the CHORD [23] study: 50 bp–1 kb, 1–10 kb, 10–100 kb, 100 kb–1 Mb, 1–10 Mb, and > 10 Mb. We calculated the Spearman correlation between the number of duplications or deletions per sample in a specific size range, and the signature exposures.

Comparison to other HR deficiency scores

We compared our multimodal signatures to other popular HRd scores/mutational signatures. First, CHORD [23] is a machine learning-based tool that detects homologous recombination (HR) deficiency in tumors using somatic mutation patterns. We obtained the CHORD probabilities p_BRCA1, p_BRCA2, and p_HRD for the PCAWG and Hartwig cohorts of ovarian cancer available in their publication ($n = 182$). For the association with HRd analysis, Hartwig samples were excluded, as their *BRCA1/BRCA2* status labels were used for training the CHORD classifier. Next, HRDetect [34] is a computational tool that predicts HRd in cancer based on SBS, ID, and rearrangement mutational signatures. We obtained their predictor probabilities for the ovarian cancer OV-AU project samples from PCAWG ($n = 68$). Moreover, the FDA-approved method for detecting HRd in the clinics, in the context of ovarian cancer is the MyChoice CDx from Myriad genetics. This is a tool that is heavily based on the analysis of a genetic instability score (GIS). This score is the sum of three values: the number of loss of heterozygosity (LOH) regions exceeding 15 Mb in size but less than the whole chromosome; the number of large-scale state transitions (LST), which are the chromosomal breaks between adjacent regions of at least 10 Mb, and the telomeric allelic imbalance (TAI), which is the number of allelic imbalances that extend to the telomeric end of a chromosome. We obtained the GIS scores for the PCAWG ovarian cancer samples [48] ($n = 108$). Finally, the CN Signature 3, reported by Macintyre et al. (2018), was established to be associated with HRd [43]. We obtained these signature exposures from their publication for the ovarian cancer OV-US project samples from PCAWG ($n = 40$).

We performed univariate Cox regressions with these scores, adding age and stage as covariates to the regression. The scores/signature exposures were used both as a continuous and as a binarized variable (splitting samples by the score mean). To compare the predictive ability

of the scores and our mutational signatures, we used the same set of samples for both survival analyses. Although mutational signatures were available for a larger number of samples, we restricted the analysis to those for which score data were also available.

CNA survival analysis

We plotted Kaplan–Meier curves for CNA of three genes known to be associated with survival in ovarian cancer patients: *AKT2*, *MYC*, and *CCNE1* [49–51]. We did this analysis for two cohorts of ovarian cancer, analyzed individually: DECIDER and PCAWG. We used tumor total copy number from different genomic regions to obtain the CNA states for each gene. The different states represent the following: 0: homozygous deletion (2-copy loss); 1: heterozygous deletion (1-copy loss); 2: normal diploid state; 3: 1-copy gain; 4: amplification (= 2-copy gain); 5: amplification (> 2-copy gain). We detected the state of the genes by using *GRanges* package and overlapping the CNA data with the genomic positions of the genes. We divided the samples into two groups: one for sample loss and neutral for a gene and other for samples with gene gains. With this classification, we plotted the Kaplan–Meier curves and calculated their median survival as previously done for mutational signatures (see section “Survival analysis”).

Signature extraction robustness test and principal component analysis

We repeated the signature extraction process with SigProfiler Extractor [46]. The number of bootstrap iterations was set to 1, in order to add variation into the result of each extraction process. The minimum and maximum number of extracted signatures was set to k . This extraction process with 1 bootstrapping run in SigProfiler was repeated 100 times. The resulting 100 S matrices, each containing the spectra of k signatures, were merged, and a principal component analysis (PCA) was performed. This analysis was done for both standard and multimodal signatures, and it was performed for the PCAWG ovarian cancer dataset.

The extracted signatures were colored in the PCA plot based on the COSMIC SBS signature contributing the highest percentage to each de novo signature. The composition percentages of these signatures were determined using SigProfiler (see “Extraction of Mutational Signatures: SigProfiler” section).

The refitting approach to estimate SBS3a and SBS3b levels

Two refitting tools were compared: SigProfiler Assignment [47] and MuSiCal [52]. The refitting approach was used to extract mutational signatures from each cohort

of ovarian cancer individually (DECIDER, Hartwig, PCAWG and OVCARE). We created a catalog of signatures by selecting a set from the de novo SBS + ID signatures obtained from the pan-HGSOC cancer analysis (all 4 HGSOC cohorts). The final set was composed of 9 multimodal signatures, representative of the most common signatures found in HGSOC: m-SBS44, m-SBS8, m-SBS5, m-SBS5 + SBS1, m-SBS3b, m-SBS3a, m-SBS13 + SBS2, and m-SBS31. Additionally, we repeated the refitting process substituting m-SBS3a and m-SBS3b for m-SBS3ab.

We used the same mutation count M matrix generated by SigProfiler MatrixGenerator [45], as input for both refitting tools. For refitting signatures with SigProfiler, the *cosmic_fit* function from SigProfilerAssignment was used with the default parameters. In the case of the MuSiCal tool, the *musical.refit.refit()* function was employed. Two different algorithms are available for MuSiCal refitting: naïve non-negative least squares (NNLS) or likelihood-based sparse NNLS. Both algorithms were run with the default parameters. The threshold parameter (“thresh”) was set to 0 for naïve and to 0.001 for likelihood-based sparse NNLS. Results from both MuSiCal refitting algorithms were broadly consistent, and the results from the likelihood-based sparse NNLS algorithm were analyzed further because of a higher Spearman correlation coefficients between the activity of the refitted signatures m-SBS3a and m-SBS3b, and the de novo signatures. For the subsequent analyses of the refitted signatures (association with HRd, survival studies, etc.) from both SigProfiler and MuSiCal, the same procedures applied to the de novo signatures were followed (see “Survival analysis” and “Association between signature activities and HR status” sections).

Association between signature activities and HR status

We studied the association between signature activity and HRd. To detect HR deficient samples, we examined both the germline and somatic mutations. In the case of germline mutations, for Hartwig and PCAWG cohorts, we took the samples with rare germline damaging variants in genes *BRCA1* and *BRCA2*, defined as previously [25]. In brief, this included truncating variants or missense variants with CADD score ≥ 25 . Moreover, variants with a population frequency higher than 1 % were removed. For samples in the OVCARE cohort, *BRCA1* and *BRCA2* deficiencies were annotated according to their *BRCA1/BRCA2* germline mutations or methylation status, as provided in their clinical metadata [40]. For samples in the DECIDER cohort, *BRCA1* and *BRCA2* both germline and somatic mutations were taken into account, as per their clinical metadata.

Moreover, we studied the somatic SNVs, indels, and homozygous deletions in the *BRCA1* and *BRCA2* genes,

as well as other HR-related genes (*ARID1A*, *ATM*, *ATRX*, *BAP1*, *BARD1*, *BLM*, *BRCA1/BRCA2*, *BRIPI*, *CHEK1/2*, *FANCA*, *MRE11A*, *NBN*, *PALB2*, *RAD50*, *RAD51*, *RAD51B*, *WRN*). For PCAWG, we used the available SnpEff annotations, whereas for DECIDER, HARTWIG, and OVCARE, we annotated VCF files using the Variant Effect Predictor (VEP) tool. We classified SNVs with the labels “stop_gained,” “missense_variant,” and “splice_region_variant,” and indels with the labels “disruptive_inframe_deletion” and “frameshift_variant.” For homozygous deletions, we used AnnotSV for annotation and included deletions with “Overlapped_CDS_percent > 80.”

We created four different HRd groups according to whether they were *BRCA1*, *BRCA2*, or other HR-related genes (listed above) deficient, or HR proficient, when no somatic or germline damaging mutations were detected in the sample for the HR-related genes. In the four ovarian cancer cohorts, around 52–66 % of patients were labeled as HR proficient, consistent with previous reports [19, 20].

Note that, because the other HR gene group is quite permissive in the sense of encompassing many different genes, of which some genes may have only distant links with HR, for this “other HR genes” group we considered only the somatic mutations and CNA; the “*BRCA1*” and “*BRCA2*” group consider the somatic and also the rare germline variants in the *BRCA1* and *BRCA2* genes.

To compare the activity of mutational signatures between *BRCA1* deficient, *BRCA2* deficient, other HR genes deficient, and remaining (putatively HR proficient) patients, we employed the Kruskal–Wallis test for significant differences in signature activities across the three groups. Next, comparisons between the groups were conducted using the Wilcoxon rank-sum test, with Bonferroni correction applied to adjust for multiple testing.

We performed this analysis on mutational signatures extracted from the pan-HGSOC cohort, as well as from individual HGSOC, prostate, TN breast, and pancreas cohorts. For signatures extracted from more than one individual cohort (all except the pan-HGSOC signatures), the HRd association analysis was conducted by combining samples from the respective cohorts.

DNA replication time (RT)-aware mutational signatures

We designed RT-aware signatures, both multimodal and standard. To do so, we used Repli-seq [53] data to delimit two RT bins. The average RT was calculated for each 200-kb window in the genome. We classified each window into two equally sized groups based on their average RT: early-RT and late-RT. Then, mutations were divided into those that occurred in early-RT and in late-RT bins.

Next, SigProfiler MatrixGenerator [45] was used to obtain the mutation count matrix *M* for each cohort, using as input the early-RT and the late-RT mutation *vcf* files separately. Therefore, for each cohort, we obtained two *M* matrices, one for early and another for late-RT mutations. These two matrices were joined. In the case of RT-standard signatures, the early-RT-SBS and late-RT-SBS matrices were merged to form a 192 mutation type matrix: 96 (early SBS) + 96 (late SBS). To obtain RT-multimodal signatures, the early-RT-SBS + ID and late-RT-SBS + ID matrices were merged to form a matrix with 358 mutation types: 96 (early SBS) + 83 (early ID) + 96 (late SBS) + 83 (late ID).

These matrices were used as input for SigProfiler Extractor for de novo extraction of mutational signatures in the same manner as previously described for non-RT-aware standard and multimodal signatures. Finally, in the subsequent analysis of these signatures, the same procedures as used for the rest of the signatures were applied. We performed the extraction of RT-aware signatures for Hartwig and PCAWG cohorts of HGSOC individually.

Results

Multimodal signature extraction of SBS3 in ovarian cancer by combining SNVs and indels

Our aim was to test the benefits of our multimodal signature extraction method for predicting prognosis in ovarian cancer and other cancer types. This approach relies on the standard NMF methods (here, SigProfiler [45]). However, it includes spectra from several types of mutations in the same extraction process (SBSs and IDs in this case) (Fig. 1a), instead of extracting signatures for each type of mutation independently. Thus, the multimodal signatures include these 96 channels for the SBS, as standard, along with 83 additional channels representing each subtype of ID mutations, for 179 channels in total (Fig. 1a).

To test our method, we used WGS samples from four independent ovarian cancer cohorts: DECIDER [37], PCAWG [38] (OV-AU and OV-US datasets), OVCARE [40], and Hartwig Medical Foundation [39] (henceforth, Hartwig), selecting the HGSOC subtype (see Methods). Considering all $n = 386$ WGS samples from these 4 cohorts in a joint, pan-HGSOC analysis, we extracted standard (unimodal) SBS-only and ID-only signatures, and multimodal signatures (with joint NMF inference over SBS and ID spectrum) and compared the results. We extracted 14 standard SBS signatures, 11 standard ID signatures, and 20 multimodal SBS + ID signatures from the pan-HGSOC cohort. Among all the extracted mutational signatures, we focused on signatures SBS3, SBS8, and SBS39, which are present among ovarian

cancer samples and can be confounded in analyses. As regards mechanisms, SBS3 was attributed to homologous recombination deficiency (HRd) as one mechanism that can generate it; SBS8 is thought to be caused by NER deficiency [10, 54], and SBS39 is an unknown signature that has been described as a possible artifact. In the case of ID signatures, we focused on signatures ID6 and ID8, also associated with HRd (see signature spectra in Additional File 2: Table S1-S3).

We identified these three SBS signatures (SBS3, SBS8, and SBS39) among the set of extracted signatures when using either the standard SBS and multimodal methods. The latter obtained two distinct SBS3 multimodal signatures (m-SBS3). The first signature combined the SNV spectrum of SBS3 with an indel spectrum resembling known ID8; we named it m-SBS3a. The second exhibited SBS3 combined with the ID6 spectrum, and was named m-SBS3b. Signature m-SBS3b more closely matched the SBS3 spectrum of SNV mutations than m-SBS3a did, which corresponded to a mix of SBS3 and SBS8 in similar proportions (Fig. 1b). Notably, the ID channels had a significant contribution to these two m-SBS3a and 3b signatures, making up 15 % and 30 % of their respective spectra (Fig. 1b). Both ID6 and ID8 spectra contain an excess of short deletions with flanking microhomology, with lengths usually ≥ 2 nucleotides (nt) in ID6, and usually shorter, at 1 nt, in the ID8 spectrum (Additional File 1: Fig. Additional File 1: Fig. S1). In contrast, for the multimodal m-SBS8 and m-SBS39, the contribution of the ID spectrum was minor: 3 % and 2 %, respectively (Fig. 1b). Multimodal signature m-SBS8 was found mainly with ID9 (Fig. 1b). A modest correlation between these two signatures was reported in a study of pediatric cancers [55]. ID9 is present across many cancer types and has an unknown etiology. As for the standard (unimodal) ID signatures, from the pan-HGSOC data we extracted both ID6 and ID8 signatures separately, which we refer to as s-ID6 and s-ID8. Multimodal signatures m-SBS3b and m-SBS3a showed a strong correlation to s-ID6 and s-ID8, respectively, with Spearman correlations of 0.96 and 0.88 (Additional File 1: Fig. S2), consistent with the ID spectrum being highly relevant to the multimodal signatures, especially m-SBS3b. Additional File 1: Fig. S2 also shows the Spearman correlations for the per-sample activities (“exposures”) for all multimodal and standard mutational signatures obtained from the pan-HGSOC cohort.

Multimodal signatures differently associate with BRCA1 versus BRCA2 deficiency

To clarify the mechanisms underlying our signatures of interest, we tested their association with damaging

germline and/or somatic variants in *BRCA1*, *BRCA2*, and somatic variants in a set of other HR-associated genes (listed in “Methods” section). As shown in Fig. 1c, we found a strong association between signature m-SBS3b and *BRCA1*- and *BRCA2*-deficient patients compared to patients with no genetic deficiencies in HR genes (“HR proficient”), with median m-SBS3b signature activity being 843, 1714, 180, and 111, respectively. The differences in signature activity were significant for both *BRCA1* and *BRCA2* genes, compared to the HR proficient group ($p = 1.1e - 06$ and $p = 6.8e - 06$, respectively), but not for the other HR genes (Fig. 1c). The change in exposure is more marked for *BRCA2* over *BRCA1*. In fact, the ratio of median exposures between *BRCA2* and HR proficient group was 15.4, and for *BRCA1* this ratio was less than half, 7.6 (see signature exposures in Additional File 2: Table S4).

Conversely, the signature m-SBS3a was significantly linked to *BRCA1* variants but not *BRCA2* variants. Patients with *BRCA1* impairment had a median m-SBS3a signature activity of 1241, compared to 441 for *BRCA2* deficient, 489 for other HR genes, and 198 for HR proficient patients ($p = 0.001$, $p = 0.004$, and $p = 2.6e - 08$, respectively) (Fig. 1c). The ratio of median exposures between *BRCA1* and HR proficient group was 6.3, while the same for *BRCA2* was only 2.3. Signature m-SBS3a was also significantly associated with other HR gene variants ($p = 0.036$), but the effect size was much smaller than for *BRCA1*.

In contrast to multimodal SBS3a and SBS3b, the standard (unimodal) signature s-SBS3 showed a significant association with *BRCA1*, *BRCA2*, and to other HR genes damaging mutations, with a median exposure of 1744, 1742, and 1368, respectively, compared to HR proficient patients whose median activity was 450. While significantly different from the HR proficient for the three HR deficient groups ($p = 5.9e - 07$ for *BRCA1*, $p = 1.65e - 05$ for *BRCA2* and $p = 0.006$ for other HR genes), the increased link of multimodal signatures m-SBS3a and m-SBS3b to *BRCA1/BRCA2* damaging mutations was stronger than for the standard signature s-SBS3 (Fig. 1c). For standard signatures, the ratio of median exposures between *BRCA1* and *BRCA2*, and the HR proficient groups were 3.87. This suggests the multimodal signatures for SBS3 better recapitulate HRd, and moreover, the distinction between m-SBS3a and m-SBS3b can give us information about the *BRCA1*-type or *BRCA2*-type of the HR impairment (Fig. 1d).

The signature SBS8, not thought to be HRd related but often confounded with SBS3 spectrum, did not show a link to *BRCA1/BRCA2*/other HR gene variants in s-SBS8 and only a slight association with *BRCA2* and other HR genes in m-SBS8 ($p = 0.006$ and $p = 0.029$),

considerably weaker than associations of m-SBS3a/3b signatures (Fig. 1c). Finally, the SBS39 signature showed a weak increase in activity in the *BRCA2* deficiencies for the s-SBS39 ($p = 0.016$), and for *BRCA1*, *BRCA2*, and other HR gene deficiency groups in case of m-SBS39 ($p = 0.00076$, $p = 0.002$, and $p = 0.025$), but this tendency is weaker than for the m-SBS3 signatures (Fig. 1c). This could be because ID6 microhomology deletions compose 15 % of the ID partition of the multimodal signature m-SBS39; however, the admixture of ID9 as well as SBS37 (Fig. 1b) argues that our SBS39 spectrum is not a bona fide HRd signature.

The indel-only signature s-ID6 similarly correlated to *BRCA1* and *BRCA2* variants compared to the HR proficient group ($p = 6.18e - 06$ and $p = 2.18e - 06$) (Additional File 1: Fig. S3), similarly to m-SBS3b. Indel signature s-ID8 showed an increased activity for *BRCA1* and other HR gene variants compared to the HR proficient group ($p = 2.12e - 08$ and $p = 0.047$), similarly to m-SBS3a. This underscores the relevance of the contribution of the ID part of the spectrum towards the multimodal signature extraction.

Comparisons with other genomic signatures of HR deficiency

Mutational signatures m-SBS3a and m-SBS3b differentially associate with *BRCA1* and *BRCA2* deficiencies. There are several described tools/mutational signatures that have been reported to also do this. We systematically compared our multimodal mutational signatures m-SBS3a and m-SBS3b, with existing HRd classifiers, including CHORD [23] (data for $n = 65$ samples available that overlapped our pan-HGSOC cohort, but were not used in CHORD Random Forest training), HRDetect [34] ($n = 68$), CN Signature 3 [43] (an HRd related signature, from Macintyre et al., 2018) ($n = 40$), and MyChoice CDx HRd test/genomic instability score (GIS) [48] score ($n = 108$). We assessed how well each method differentiated HRd groups. Our multimodal mutational signatures m-SBS3a and m-SBS3b, as well as the sum of both (m-SBS3a + m-SBS3b), outperformed HRDetect, GIS scores, and the CN 3 signature exposures in their association with HRd gene status as defined here (Additional File 1: Fig. S4 shows results of statistical tests). The probabilities of the CHORD classifier have shown a very robust association with the HR gene deficiencies (Additional File 1: Fig. S4), nearly matching the multimodal mutational signatures. We note that CHORD and HRDetect are supervised methods, trained specifically to differentiate between known HRd mechanisms. However, our NMF-based, fully unsupervised approach is applicable without modification to diverse mutational processes, beyond HRd.

Inspired by the CHORD publication [23], we directly analyzed the Spearman correlation between mutational signature activity and the number of different sized duplications (Additional File 1: Fig. S5). The four mutational signatures show significant correlations with duplication sizes 50 bp to 100 kb. The correlations with duplications were stronger for the *BRCA1* associated signatures, m-SBS3a and s-ID8, than for the *BRCA2* associated signatures m-SBS3b and s-ID6. This result is in line with the CHORD paper findings, which showed that duplications had a bigger importance for predicting the *BRCA1*-type of HRd. The biggest difference between signatures is observed for 1–10 kb duplications, where m-SBS3a and s-ID8 exhibit a Spearman correlation of 0.68 and 0.66, respectively, while these values were of 0.46 for m-SBS3b and 0.49 for s-ID6 (all $p < 0.001$). Again, this result coincides with 1–10 kb duplications being the second most important feature for predicting *BRCA1* deficiency in the CHORD tool.

Moreover, CHORD [23] and Funnell et al. reported the association between small deletion structural variants (SVs) (50 bp–1 kb) and HRd (particularly *BRCA2* deficiency) [56]. Therefore, we also assessed the correlation between mutational signature exposures and the number of deletions by size. As shown in Additional File 1: Fig. S7, *BRCA2* associated signatures m-SBS3b and s-ID6 displayed strong correlations with 50 bp–1 kb deletions (0.88 and 0.92, $p < 0.001$), while *BRCA1* associated signatures showed slightly lower values (0.80 and 0.86, $p < 0.001$).

Multimodal mutational signatures as predictive markers of cancer patient survival

Next, we tested the ability of both standard SBS and ID, and multimodal SBS +ID signatures to predict overall survival in HGSOc patients. First, a Kaplan–Meier model was applied to each signature of interest, classifying patients into two groups: one considered as signature-negative (activity below threshold), and the other as signature-positive (above threshold, here implying top 30 % of samples, although different thresholds were tested).

As observed in the survival curves (Fig. 2a), expectedly [33, 57, 58] the HRd signature s-SBS3 was linked to a better overall survival, with a difference in the median survival between the signature-positive and signature-negative groups of 18.2 months ($p = 0.000125$) for HGSOc patients. Multimodal m-SBS3 signatures were substantially more informative here, with m-SBS3b-positive patients having a median survival of 31.2 months more than the negative group ($p = 1.55e - 07$). We note that this simple analysis pools together patients with different age and disease stages, disregarding potential confounding effects thereof (see analyses below). Signature

s-ID6 also predicted survival well (albeit less so than the multimodal signature) with a difference in median survival time of 30.6 months ($p = 2.31e - 06$) (Fig. 2a). The m-SBS3a was similarly predictive as the standard, unimodal s-SBS3 (17.7 months difference, $p = 0.00282$), as well as unimodal indel signature s-ID8 (15.8 months, $p = 0.00836$) (Fig. 2a). Furthermore, this result was qualitatively consistent throughout the different tested thresholds (Additional File 1: Fig. S7).

For comparison, we considered the standard and multimodal SBS8 and SBS39 (Additional File 1: Fig. S8). Neither s- nor m-SBS39 were significantly linked to longer survival at any threshold tested, supporting the absence of an association between SBS39 and HRd. However, s-SBS8 and the (presumably less admixed) m-SBS8 did show a significant association with a longer overall survival, with a difference in median survival of 17.1 ($p = 0.00355$) for s-SBS8 and of 17.1 ($p = 0.0134$) for m-SBS8 (Additional File 1: Fig. S8). The SBS8 might complement multimodal SBS3 in predicting survival in HGSOc, possibly because it captures a different mechanism of DNA repair failure.

To further analyze association with overall survival, we applied a Cox regression model drawing on multiple variables. All mutational signatures, of each type, were included in the regression. Age, stage of the disease, and cohort were also included as covariates, to isolate the effects particular to individual signatures. Here, we also varied the percentile threshold used to binarize into “positive” and “negative” by the activity of the signature of our interest.

For the standard signatures, s-SBS3 was associated with a longer survival in the Cox regression controlling for age/stage/cohort, with a minimum Z-score of -3.44 across different percentile thresholds (hazard ratio (HR) = 0.52, $p < 0.001$, at 40 % threshold signature activity) (Fig. 2b; Additional File 2: Table S5). Similarly, s-SBS8 and s-SBS39 are less strongly associated, with a best Z-score of -2.39 (HR = 0.62) and -2.30 (HR = 0.72), respectively (Fig. 2b). In contrast, multimodal signature m-SBS3b is strongly and significantly linked to a higher survival probability throughout most thresholds tested (except the highest ones 70–80th percentile), with a best Z-score of -4.32 (HR = 0.42, $p < 0.001$; at the for 50 % threshold).

Additional considerations in the survival associations of SBS3 and other signatures

We observed the link of m-SBS3a to significantly better prognosis in the univariate Kaplan–Meier analysis (Fig. 2a); however, it was not observed when applying the Cox multivariate regression (Fig. 2b). This is possibly because the prognostic value of m-SBS3a levels

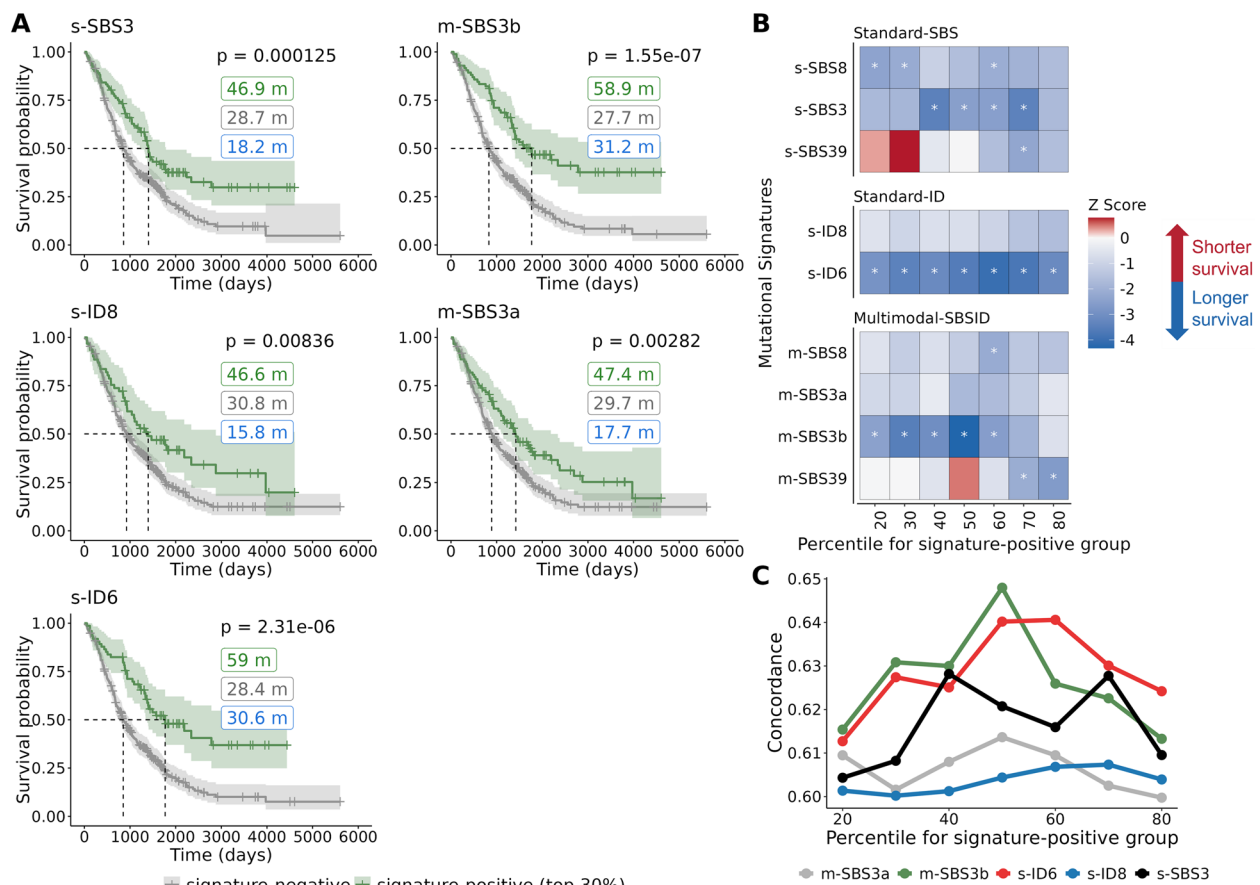


Fig. 2 Multimodal mutational signatures combining SBSs and IDs are strong predictors of survival. **A** Kaplan–Meier curves for the standard SBS signature s-SBS3, the standard ID signatures s-ID6 and s-ID8, and the multimodal signatures m-SBS3a, m-SBS3b extracted from the pan-HGSOC cohort. Samples were classified into signature-positive or signature-negative groups based on their inferred exposure (activity) to the signatures. Median survival months for each group is shown, as well as their difference colored in blue. **B** Results for multivariate Cox regression analysis for standard and multimodal HRd SBS3, ID6, ID8, and other related signatures. The x axis shows the percentile used as threshold to classify samples as signature-positive or signature-negative. Colors show the Z-scores resulting from the different Cox regressions, with negative Z-scores indicating longer survival and the positive Z-scores shorter survival. The asterisks indicate the significance of the results (adjusted p -value < 0.05), indicating whether the Z coefficient is equal to zero (no effect) or statistically different from zero. **C** Concordance comparison between m-SBS3a, m-SBS3b, s-ID6, s-ID8, and s-SBS3 signatures, obtained from univariate Cox regression using age, stage, and cohort as additional covariates. Concordance was obtained for different percentiles used as threshold to classify samples as signature-positive or signature-negative, shown in the x axis

diminishes when the analysis is adjusted by the presence of other signatures such as m-SBS3b, age, and tumor stage. We also considered the indel signature s-ID6, which also showed a consistent significant association with survival, with the lowest Z-score being -4.03 ($HR = 0.5$, $p < 0.001$) (Fig. 1b), trailing m-SBS3b as the best predictor. Signature s-ID8 did not exhibit a significant prognostic effect.

Moreover, we assessed the predictive performance of the survival models by the C-index or concordance index for the univariate Cox regressions performed individually for signatures (plus age, stage, and cohort) (Fig. 2c). The highest concordance (0.648) was achieved by multimodal m-SBS3b, at the binarization threshold

of 50 %. Signature m-SBS3b outperformed others in 4 out of 7 thresholds tested, while s-ID6 showed the highest concordance in the remaining more extreme 3 thresholds (Fig. 2c). This indicates that ID signatures do exhibit a good ability to predict overall survival, although m-SBS3b performs better when selecting a most-accurate solution overall, and is also robust across a wider range of binarization thresholds. The unimodal, “classical” SBS3 signature always trailed m-SBS3b or s-ID6 (see Fig. 5e). Moreover, m-SBS3a outperformed s-ID8 in 5 out of 7 thresholds, in line with the advantage of the multimodal approach in capturing HRd-associated survival (Fig. 2c).

The multivariate Cox regression results are shown in Additional File 1: Fig. S9 as a forest plot, for signatures SBS3, SBS39, SBS8, ID6, and ID8 at various survival thresholds, and in Additional File 1: Fig. S10 for the rest of the standard and multimodal signatures. All other mutational signatures we extracted in the pan-HGSOC analysis were weaker predictors of survival, with some signal in an unknown signature with a flat SBS spectrum and a weak component of ID9 (“m-SBSUnk-ID9” longer survival, $p < 0.05$ in the multimodal signatures; SBS spectrum is a best match with mixture of SBS5 with SBS39 although at low similarity, Additional File 1: Fig. S10). Further, there was some effect of a SBS31-like signature (shorter survival, $p < 0.05$ in the standard and multimodal signatures, see Additional File 1: Fig. S10). We note SBS31 was reported to be associated with prior treatments with platinum drugs [57] and so this result is in line with the general trends observed in cancer cell line genomes, where mutational signatures of prior exposures to DNA damaging agents commonly predict resistance to therapies [12]. Finally, for the standard (unimodal) ID signatures, no signature apart from s-ID6 exhibited a significant survival association.

We conducted Kaplan–Meier analyses to further investigate why age and stage did not appear as significant predictors in the multivariable regression (Additional File 1: Fig. S11). Given the differences in age distribution across cohorts, we calculated age tertiles within each cohort. Age tertiles showed near-significant survival differences ($p = 0.0552$), and stage significantly influenced survival ($p = 0.0223$) (Additional File 1: Fig. S11). Multicollinearity was unlikely to present an issue (see Methods). In addition to overall survival, which can be impacted by several external factors, we also evaluated the association of our multimodal mutational signatures, m-SBS3a and m-SBS3b, with progression-free survival (PFS). Again, we performed Cox regression analysis with m-SBS3a and m-SBS3b individually, and age, stage, and cohort as covariates for DECIDER, PCAWG OV-US, and OVCARE cohorts, where we had PFS data available. Both multimodal mutational signatures SBS3a and SBS3b show a significant association with longer PFS, just as with longer overall survival, without strong differences between the two survival metrics (Additional File 1: Fig. S12). Overall survival prediction does exhibit a higher concordance index for both multimodal signatures, compared to when predicting PFS (Additional File 1: Fig. S12). Therefore, our multimodal mutational signatures can predict both overall survival (as previously shown) and PFS.

Complementarity between mutational signatures in predicting cancer survival

To assess putative redundancies between the various signatures, we asked if these mutational signatures were predictive of overall survival on their own or in

combination with each other. We ran a Cox regression analysis using different signature combinations, as well as these signatures on their own, adjusting for patient age, tumor stage, and cohort. We did this for standard and multimodal signatures (Additional File 1: Fig. S13).

Neither the standard nor the multimodal SBS39 signature is predictive of overall survival in ovarian cancer patients for any combination of signatures, only m-SBS39 when added individually ($Z = -1.98$, $HR = 0.70$). Interestingly, when SBS8 was included alone in the analysis, it was associated with $Z = -2.78$ ($HR = 0.67$) for m-SBS8, and $Z = -3.35$ ($HR = 0.62$) for s-SBS8 (Additional File 1: Fig. S13). However, when s-SBS8 was included with s-SBS3, and m-SBS8 with m-SBS3b, SBS8 exhibited a lower predictive ability ($Z = -2.01$, $HR = 0.73$) and was not significantly predictive of survival, respectively. This suggests that SBS8 is redundant with SBS3, rather than complementary to it, and so it does not make for a promising survival marker in the context of m-SBS3b.

Regarding m-SBS3a, which was significant if included alone in the regression ($Z = -2.75$, $HR = 0.65$), it did maintain its prognostic value when including m-SBS8 and m-SBS39 to the regression (Additional File 1: Fig. S13). However, when m-SBS3b was added alongside m-SBS3a to the Cox regression, its predictive capacity was diminished and not significant ($Z = -0.78$, $HR = 0.88$), suggesting that m-SBS3a is redundant for predicting survival if m-SBS3b is considered. Similarly, s-ID8 was also significant when added alone ($Z = -2.64$, $HR = 0.62$), but not when s-ID6 was also included.

S-SBS3, s-ID6, and m-SBS3b showed the strongest predictive ability and maintained a significant link with higher survival probability throughout all the tests. The Z-scores for s-SBS3 varied between -3.61 and -2.43 , depending on which other signatures were added. The Z-scores for s-ID6 ranged from -4.61 to -4.05 . Finally, the Z-scores for m-SBS3b ranged from -4.99 (best score accomplished when it is applied alone), to -3.86 , when adjusted by m-SBS3a, m-SBS39, and m-SBS8 in the same regression.

In conclusion, our multimodal method allows the extraction of two signatures caused by HRd: m-SBS3a, very strongly linked to *BRCA1* damaging variants, and m-SBS3b, associated with both *BRCA1* and *BRCA2* impairment, more so the latter. Multimodal SBS3b has the strongest association with a longer survival, while the other SBS or multimodal signatures have a weaker effect that is not robustly seen upon adjusting for m-SBS3b exposure.

Testing consistency of multimodal SBS3 signature in individual cohorts

We asked if the same results for HRd and survival association replicated within each ovarian cancer study individually. Because the mutational signature extraction is

performed from a lower number of WGS samples (here, DECIDER $n = 100$, PCAWG $n = 110$, Hartwig $n = 124$, OVCARE $n = 52$), fewer mutational signatures emerged from each cohort: between 4 and 8 for s-SBS, and between 7 and 9 for m-SBS + ID (see signature spectra in Additional File 2: Table S1–S3). We detected inconsistencies in being able to identify both signatures SBS3 and SBS8 in every cohort when using the standard, unimodal SBS methodology (i.e., using SigProfiler on SBS spectra only) (Additional File 1: Fig. S14a), even though SBS3 is expected to always be present in ovarian cancers, which are fairly commonly HR deficient.

Next, we compared this for the standard and the multimodal method (Additional File 1: Fig. S14a), shown for the number of signatures extracted (k) recommended by SigProfiler, and additionally for $k + 1$ and $k - 1$ (total of 4 studies, and 3 k settings each). For the unimodal signatures, s-SBS8 is always extracted but s-SBS3 is missing in 5/12 cases. When considering only the k recommended by SigProfiler (ignoring $k + 1$ and $k - 1$), SBS3 is only identified for 2/4 ovarian cancer cohorts (Additional File 1: Fig. S14a). Signature s-SBS39 was also missing 7/12 times. We hypothesize that the failure to extract SBS3 and SBS39 in the unimodal setup is because of the “flatness” of their spectra, and that this issue is aggravated when the number of analyzed samples is smaller. This was previously described for these and for other “flat” signatures, such as the widespread SBS5 and SBS40 [2].

However, the multimodal approach using SigProfiler allowed a consistent extraction of both SBS8 and SBS3 in each of the four cohorts (Additional File 1: Fig. S14a), suggesting that joint analysis of SBS and IDs helps extract known mutational signatures in small cohorts. In this analysis of smaller-sized, individual cohorts, we could not distinguish signatures m-SBS3a and m-SBS3b, but instead recovered a combination: signature SBS3 + ID6 + ID8, here named m-SBS3ab. Signature m-SBS3ab was very consistent across cohorts (spectra shown in Additional File 1: Fig. S15) with cosine similarities between them ranging from 0.96 to 0.99. In every cohort, the contribution of ID6 was higher than ID8 in the ID channels of the m-SBS3ab signatures—it ranged from 58.2 to 61.8%—suggesting this multimodal signature does more strongly reflect signal from m-SBS3b of the pan-HGSOC analysis above.

The multimodal method produces an m-SBS3ab signature with a higher mean cosine similarity to the COSMIC reference SBS3, 0.952, compared to the standard SBS3 extraction, with a cosine similarity of 0.935 ($p = 0.024$). This suggests that the inclusion of IDs improved the SBS component of the spectrum (Additional File 1: Fig. S14b). Additionally, the standard s-SBS3 signatures were primarily matched to a mixture of SBS3 and SBS39, with SBS3 accounting for 60–64 % of the spectrum. In

contrast, the m-SBS3ab signatures showed an increased proportion of SBS3, ranging from 81 to 94 % of the SBS component, except in the case of Hartwig, where SBS3 was combined with SBS40.

We next considered the association between signature activity and the presence of damaging germline and/or somatic variants in *BRCA1*, *BRCA2*, and other HR genes (listed in Methods section) (Additional File 1: Fig. S14d). For the standard s-SBS8 levels, we observe a modest association with *BRCA2* ($p = 0.003$), *BRCA1* ($p = 0.00069$), and other HR gene ($p = 0.012$) variants (see signature exposures in Additional File 2: Table S6–S9). Previous reports showed that signature SBS8 did correlate with *BRCA1/BRCA2* deficiency, but less so than SBS3 [26, 59, 60], suggesting that the association of SBS8 to *BRCA1/BRCA2* may have resulted because the spectrum is difficult to unmix from SBS3. Also, in our ovarian WGS analysis, the standard s-SBS8 and s-SBS3 both associate with *BRCA1/BRCA2* variants (Additional File 1: Fig. S14d). Mechanistically, SBS8 is likely to be caused by a NER pathway deficiency [10, 61], arguing against a link of SBS8 with HRd. Indeed, our multimodal signature analysis confirms this. The association between SBS8 and *BRCA1/BRCA2* deficiency is not observed for the m-SBS8. In the same vein, for SBS3, the multimodal m-SBS3ab signature shows improvement over s-SBS3 in the link for *BRCA1* ($p = 1.79e - 07$), and *BRCA2* ($p = 2.95e - 05$) gene defects, with a median signature exposure of 2840 and 3876 (mean across cohorts), respectively, compared to the median activity of 303 for the HR proficient group (Additional File 1: Fig. S14d). Thus the ratio of median m-SBS3ab signature activity between *BRCA1* and *BRCA2*-variant groups, and the HR proficient group was 9.37 and 12.79, respectively, while for s-SBS3 the same ratios were 2.49 and 2.82, respectively. ID signature s-ID6 + ID8 also significantly correlated to *BRCA1* and *BRCA2* defects ($p = 6.54e - 07$ and $p = 4.75e - 06$, respectively), even though the median activity ratios to the HR proficient group were smaller for the ID-only signature than for m-SBS3ab: 7.42 for *BRCA1* and 11.28 for *BRCA2*. Thus, m-SBS3ab is more reliable for detecting HRd than ID signatures alone.

These results are consistent when examining each cohort individually. Specifically, the ratio of median m-SBS3ab signature activity for the *BRCA1* group compared to the HR proficient group in DECIDER is 8.14, in Hartwig is 15.24, in PCAWG is 5.58, and in OVCARE is 10.15. For the *BRCA2* group, the ratios are 1.66, 15.40, 10.76, and 13.73, for each cohort, respectively. DECIDER exhibits the lowest ratio, which is explained by 3 samples (two with somatic variants and one with germline mutation in *BRCA2*) that show a signature activity lower than 200. Barring this exception, the result is observed

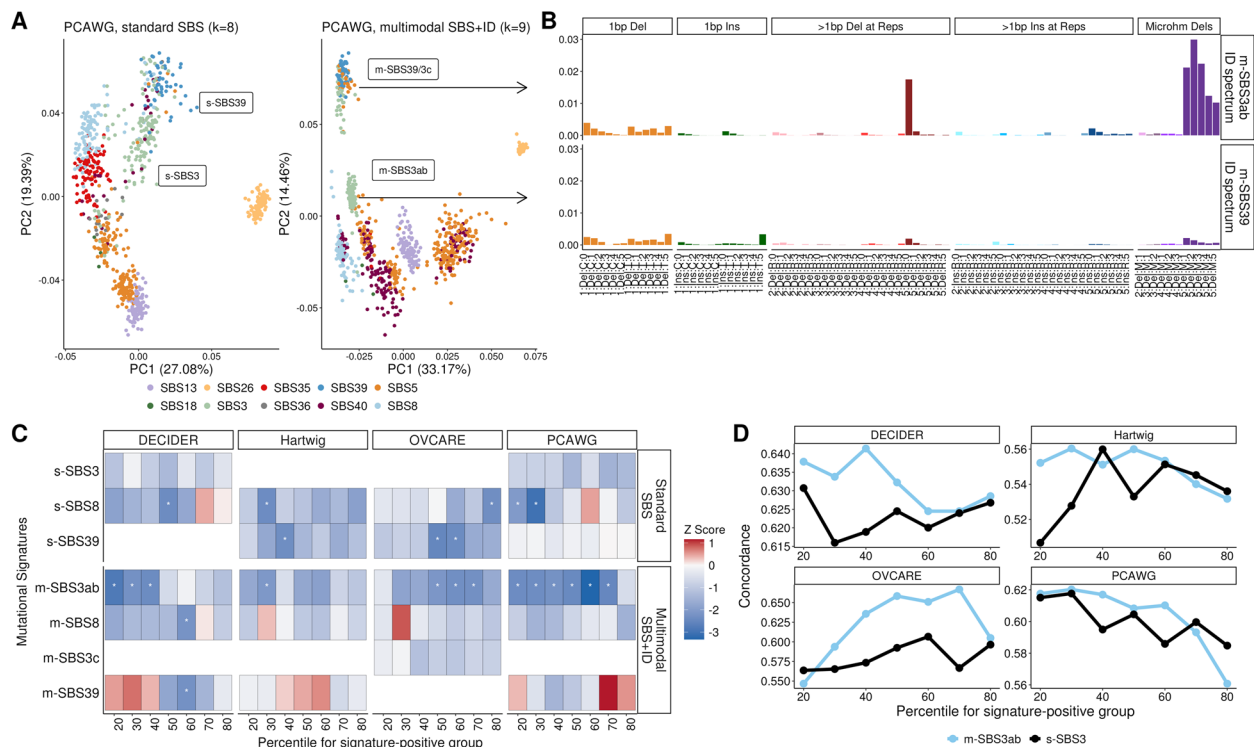


Fig. 3 Analysis of multimodal SBS + ID mutational signatures of HRd in four cohorts of ovarian cancer. **A** PCA of standard and multimodal signatures for the PCAWG cohort, obtained from 100 repetitions of the extraction processes performed on bootstrapped data. Colors denote the COSMIC reference signature with the highest contribution to the SBS spectrum. **B** ID spectrum for m-SBS3ab and m-SBS39/SBS3c clusters shown in the PCA. **C** Results of multivariate Cox regression comparing standard and multimodal signatures in their association with overall survival. The x axis shows the different percentiles tested to classify samples by signature activity, and the asterisks represent a significant association between the variable and overall survival ($p < 0.05$). **D** Concordance comparison between m-SBS3ab and s-SBS3 for each individual cohort, obtained from univariate Cox regression using age and stage as additional covariates. Concordance was obtained for different percentiles used as threshold to classify samples as signature-positive or signature-negative, shown in the x axis

independently in 4 separate cohorts, and it aligns with the above results in the pan-HGSOC cancer analysis, wherein multimodal approach clarifies the association between HRd and SBS3 spectrum.

In the multimodal signature analyses, an additional SBS3-like signature was identified; we suggest this m-SBS3c may be another manifestation of the same process that generates SBS39, which is less likely to result from HRd (see Additional File 3: Text S1 for details, and Additional File 1: Fig. S14a, c, d). We further study the stochastic variation across NMF extraction runs (Fig. 3a,b), and across cohorts (Fig. 3c), showing how in unimodal analysis SBS39 and SBS3 are mixed, but applying the multimodal approach unmixes them, based on distinct indel spectra (see Additional File 3: Text S1).

Survival analyses in individual studies and comparisons to established copy number markers

Next, we conducted a multivariate Cox regression to study the relationship to these various mutational signatures with overall survival for each individual cohort

of ovarian cancer. This revealed significant associations between the standard (SBS-only) signature s-SBS8 and s-SBS39 to longer survival, with the best Z-score across 4 cohorts of -3.02 ($HR = 0.36$, $p = 0.003$) and -2.51 ($HR = 0.40$, $p = 0.012$), while the association for signature s-SBS3 and its best Z-score of -1.51 ($HR = 0.59$, $p = 0.132$) were weaker (Fig. 3c).

However, with multimodal signatures, we observed a strong and consistent association between m-SBS3ab and higher survival probability across the four cohorts considered individually, and many different thresholds tested (Fig. 3c). The best Z-scores and respective HR and p -values were as follows: -2.83 ($HR = 0.28$, $p = 0.005$) for DECIDER, -3.33 ($HR = 0.37$, $p < 0.001$) for PCAWG, -2.11 ($HR = 0.53$, $p = 0.034$) for Hartwig, and -2.32 ($HR = 0.38$, $p = 0.02$) for OVCARE.

The effect on survival of multimodal m-SBS8 and m-SBS39 was weaker than their standard version, and significant in just a few of the multiple thresholds and cohorts tested (Fig. 3c). These results further suggest that our multimodal approach allowed for a better separation

of SBS3, SBS39, and SBS8 signatures in small cohorts, resulting in a more accurate and robust prediction of survival, which better matches the known link between HRd and prognosis in HGSOc.

Finally, we compared the concordance index obtained from univariate Cox regressions for m-SBS3ab and s-SBS3 performed for each cohort (and adjusting by age and stage of the disease) (Fig. 3d). Generally, we can observe a lower concordance for s-SBS3 than for m-SBS3ab.

We further compared the predictive ability of the multimodal m-SBS3ab in individual cohorts (corresponding to m-SBS3b from the pan-HGSOc analysis) with several established genomic markers of ovarian cancer survival, based on CNA. In particular, we considered CNA gains in *CCNE1* and in *MYC*, which were reported as associated with worse prognosis in ovarian cancer and also across other cancer types [49, 50]. We further considered CNA gains in the *AKT2* oncogene, one of the top-ranked survival predictors by CNA status and also supported by RNA-seq levels association in the TCGA ovarian cohort (survival data from Lui et al. [51] and <https://www.tcgasurvival.com/>). Considering the PCAWG cohort, indeed the CNA gains in the three genes predicted 9.2, 18.6, and 10.6 months shorter survival, for *CCNE1*, *MYC*, and *AKT2* respectively (Additional File 1: Fig. S16). For comparison, in PCAWG the high m-SBS3ab levels predicted 29.4 and 22.6 months longer survival, at 20th and 50th percentile thresholds, respectively. We further checked the DECIDER cohort of ovarian cancer, where the 3 prognostic marker gene CNA gains predicted 18.9, 0.39, and 9.5 months shorter survival (Additional File 1: Fig. S16), while the high m-SBS3ab levels predicted 38.5 and 13.3 months longer survival at the two thresholds (Additional File 1: Fig. S16). Overall, the multimodal signature m-SBS3ab strongly exceeds known genetic markers in predicting survival of HGSOc patients.

Refitting provides a practical yet accurate method for multimodal HRd signature inference

Because of the lower number of samples and mutations available when using each cohort separately, it becomes challenging to extract signatures m-SBS3a and m-SBS3b, whose distinction is important. A convenient way to recover these signatures would be to use a refitting approach (typically performed by a NNLS regression, as in SigProfiler [47]), instead of a de novo extraction method by running NMF. We wanted to test whether the refitting (assignment) approach would be useful for m-SBS by creating a reference set of mutational multimodal signatures to fit to, including m-SBS3a and m-SBS3b. We tested it in different ovarian cancer cohorts and compared two popular methods for mutational

signature assignment: the SigProfiler Assignment [47] and MuSiCal [52] tools.

The two outputs of both tools differ in the sparsity of the distribution of the exposures: SigProfiler assigned m-SBS3a and m-SBS3b exposures of zero in 86 % and 80 % of the ovarian tumors we analyzed, whereas MuSiCal only did so for 20 % and 38 %, respectively, more similarly to the non-zero rates in de novo NMF extraction. Next, the MuSiCal signatures exhibited a stronger correlation with the de novo NMF extracted signatures' exposures, with a mean Spearman correlation between m-SBS3b de novo and MuSiCal of 0.92, compared to 0.63 for SigProfiler refitting (Fig. 4a). Similarly, for m-SBS3a, the correlation between exposures was 0.84 with MuSiCal and 0.52 with SigProfiler, suggesting MuSiCal refitting could be a viable approach in generating prognostic marker mutational signatures (Fig. 4a).

To assess the assignments, we tested prediction of survival and association with damaging germline and/or somatic variants in *BRCA1*, and *BRCA2*, and other HR genes. Both m-SBS3a and m-SBS3b, obtained with SigProfiler and MuSiCal, showed similar associations with *BRCA1/2* mutations to those found with de novo extraction (Fig. 4b and Additional File 1: Fig. S17a). Among the signatures obtained from MuSiCal, m-SBS3a showed a strong link to only *BRCA1* defects compared to the HR proficiency group ($p = 7.159 \times 10^{-8}$). For m-SBS3b, MuSiCal signature associates with *BRCA1* and *BRCA2* defects, being more enriched in *BRCA2*. This signature showed a median exposure, by MuSiCal refitting, of 753 for *BRCA1* deficient group and 1757 for *BRCA2* deficient group, compared to a signature activity of 228 for HR proficient patients ($p = 2.59 \times 10^{-5}$ and $p = 2.29 \times 10^{-6}$, respectively). The statistics associated with the same analysis for the SigProfiler refitted signatures are shown in Additional File 1: Fig. S17a.

The ratio between *BRCA2*-deficient and HR proficient is thus 7.7 for m-SBS3b fit by MuSiCal, and it was 15.4 for m-SBS3b de novo signatures inferred by NMF. For m-SBS3a, the ratio between *BRCA1*-deficient and HR proficient groups is 3.7 for MuSiCal signature and 6.2 for de novo. Therefore, even though the associations to HRd found for MuSiCal refitted signatures are strong, the de novo signatures show even stronger results.

Next, we tested their ability to predict patient overall survival. Because of the constraints imposed by the sparseness of the SigProfiler refitting output, with approximately 80 % of the samples for signatures m-SBS3a and m-SBS3b showing null signature activity, those samples were classified as signature-negative. For MuSiCal, to be comparable, the top 20 % of samples were also labeled as signature-positive and the other 80 %, as signature-negative. The Kaplan–Meier curves for

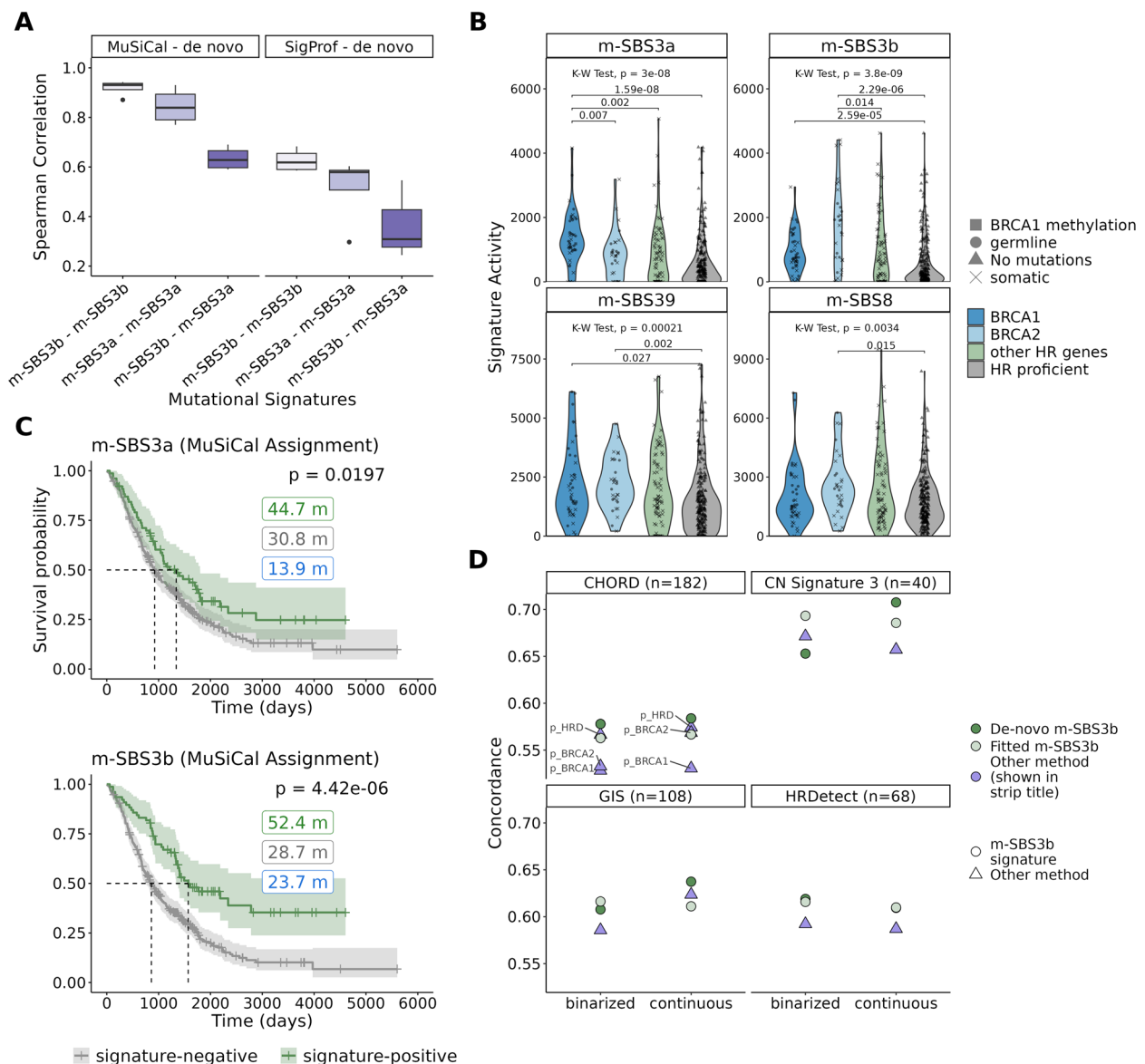


Fig. 4 Mutational signature refitting methods allow extraction of m-SBS3a and m-SBS3b. **A** Spearman correlation between signature activities (exposures) of the de novo and the assigned (refitted) signatures, using MuSiCal and SigProfiler refitting tools. **B** Association of mutational signatures obtained by refitting to pathogenic variants in *BRCA1*, *BRCA2*, or other HR genes (see gene list in Methods) damaging germline and/or somatic variants (Kruskal-Wallis test followed by a Wilcoxon rank-sum test). **C** Kaplan-Meier curves for signatures m-SBS3a and m-SBS3b extracted from four cohorts of ovarian cancer samples by MuSiCal refitting. The median survival time is shown for both groups, as well as the difference between them in blue. **D** Comparison of the concordance (C-index) obtained from Cox regression individually testing multimodal mutational signature m-SBS3b (both de novo extracted, and refitted), and four different scores for detecting HRd (CHORD, HRDetect, CN Signature 3 and GIS), with age and stage as covariates. The scores/signature exposures were used both as a continuous and as a binarized variable (obtained by dividing samples into two groups, using the score median as a threshold). Number of WGS samples with a given score being available for comparison is provided as n in the header

both methods overall agree with the results found with the de novo method (Fig. 4c and Additional File 1: Fig. S17b). For MuSiCal, the m-SBS3b signature-positive group showed a longer survival than the signature-negative group. The median survival difference between the

two classes was 23.7 months ($p = 4.42e - 06$). Multimodal signature m-SBS3a obtained by MuSiCal was also predictive of a longer overall survival, with an increase in median survival of 13.9 months, although the statistical significance was more modest than for m-SBS3b ($p =$

0.0197). Similar results were obtained from the survival analysis performed for m-SBS3a and m-SBS3b signatures extracted with SigProfiler refitting method, and Kaplan–Meier curves are shown in Additional File 1: Fig. S17b.

In essence, the refitting approach offers a quicker and simpler, yet accurate approach to detect signatures m-SBS3a and m-SBS3b in new WGS-sequenced tumor samples, potentially valuable in clinical use.

Resolved multimodal SBS3 into SBS3a and SBS3b compares favorably to other genomic HRd markers

To support that the joint m-SBS3ab signature, recovered in individual HGSOc cohorts, can be meaningfully split into two separate m-SBS3a and m-SBS3b signatures, we tested the effect of replacing m-SBS3a and m-SBS3b with m-SBS3ab in the MuSiCal refitting tool. The resulting signature exposures of m-SBS3ab across HGSOc samples showed Spearman correlations of 0.65 with m-SBS3a and 0.87 with m-SBS3b. As for the signatures' associations with mutations in HR genes: m-SBS3a exhibited a better-supported association with *BRCA1* deficiency than m-SBS3ab did ($p = 1.59e - 08$ vs. $p = 1.19e - 06$), and m-SBS3b to *BRCA2* ($p = 2.29e - 06$ vs. $p = 1.97e - 05$) than m-SBS3ab did (Additional File 1: Fig. S18a).

Additionally, the Kaplan–Meier curves indicated that the refitted m-SBS3ab is also significantly associated with longer survival time, with a difference in median survival of 17.3 months between the signature-positive and signature-negative groups ($p = 9.835e - 05$) (Additional File 1: Fig. S18b). However, this difference is smaller than for m-SBS3b (23.7 months, Fig. 4c). We also assessed concordance from a univariate Cox regression to predict overall survival in HGSOc, for m-SBS3a, m-SBS3b, and m-SBS3ab obtained via MuSiCal refitting. While m-SBS3ab demonstrated high concordance (ranging from 0.59 to 0.631), m-SBS3b consistently achieved higher and more stable concordance values (ranging from 0.61 to 0.64 across different binarization thresholds), highlighting its improved predictive value as an HRd marker compared to m-SBS3ab (Additional File 1: Fig. S18c).

We analyzed how well our multimodal mutational signatures m-SBS3a and m-SBS3b predicted survival, compared with existing HRd classifiers, including CHORD [23] ($n = 182$ overlapping samples with our pan-HGSOc cohort), HRDetect [34] ($n = 68$), copy number (CN) Signature 3 [43] ($n = 40$), and MyChoice CDx HRd test/GIS score [48] ($n = 108$). MyChoice CDx HRd test (referred to as GIS) is the FDA-approved test to detect HR deficient patients for ovarian and breast cancer in the clinics. We evaluated survival prediction using Cox regression models (Additional File 1: Fig. S19). Both refitted and de novo m-SBS3b signatures were significantly associated with

improved survival, outperforming CHORD, where only the HRd probability reached significance (Additional File 1: Fig. S19; Additional File 2: Table S10). HRDetect also showed predictive value, though weaker than m-SBS3b. CN Signature 3 did not significantly associate with survival, while the GIS score did, but with lower predictive power than m-SBS3b (Additional File 1: Fig. S19). Concordance (C-index) analyses further supported the superiority of m-SBS3b over most other classifiers, including GIS, in the de novo form of m-SBS3b (Fig. 4d).

Therefore, the comparison of our multimodal mutational signatures to other HRd classifiers underscores the potential clinical utility of signatures m-SBS3a and m-SBS3b as biomarkers in the clinics, specifically through multimodal signature refitting due to its easier implementation.

Multimodal SBS3 predicts survival in platinum-treated patients in breast, pancreas, and prostate cancer

We repeated the same analysis, asking if SBS3 signatures predict survival, in three other cancer types known to be commonly affected by HRd: breast, pancreas, and prostate from the Hartwig cohort. In the case of breast cancer, we focused on the triple-negative (TN) subtype because of its enrichment with HR deficient patients. In these three tissues, signatures s-SBS3, s-SBS8, and s-SBS39 were also not consistently extracted when applying the standard method (Fig. 5a; see signature spectra in Additional File 2: Table S1 and S3). Sometimes, signatures were mixed together (for example, s-SBS3 + SBS39 in prostate cancer and s-SBS8 + SBS3 in breast cancer). However, these signatures were separately extracted when using the multimodal approach, except in the case of pancreatic cancer (Fig. 5a) where, atypically, SigProfiler recommended a lower number of multimodal signatures ($k = 11$) than standard signatures ($k = 15$). For $k = 15$ multimodal signatures in the pancreas cohort, we did identify m-SBS3ab, m-SBS8 and m-SBS39.

We also assessed the association of these mutational signatures to damaging germline variants in *BRCA1* and *BRCA2* genes for the three cancer types; please see Additional File 3: Text S2 for analyses and Additional File 2: Tables S11–S13 for signature exposures.

Next, we performed a survival study for these cancer types, revealing no clear correlation between SBS3, both standard and multimodal, and improved overall survival outcomes. Notably, these three cancer types are treated by diverse drugs, unlike HGSOc where platinum and taxane are first-line therapy. For TN breast cancer, we did observe significantly higher median survival times for both s-SBS8/SBS3 mixed signature and m-SBS3ab (Fig. 5c), with survival outcomes not significantly better

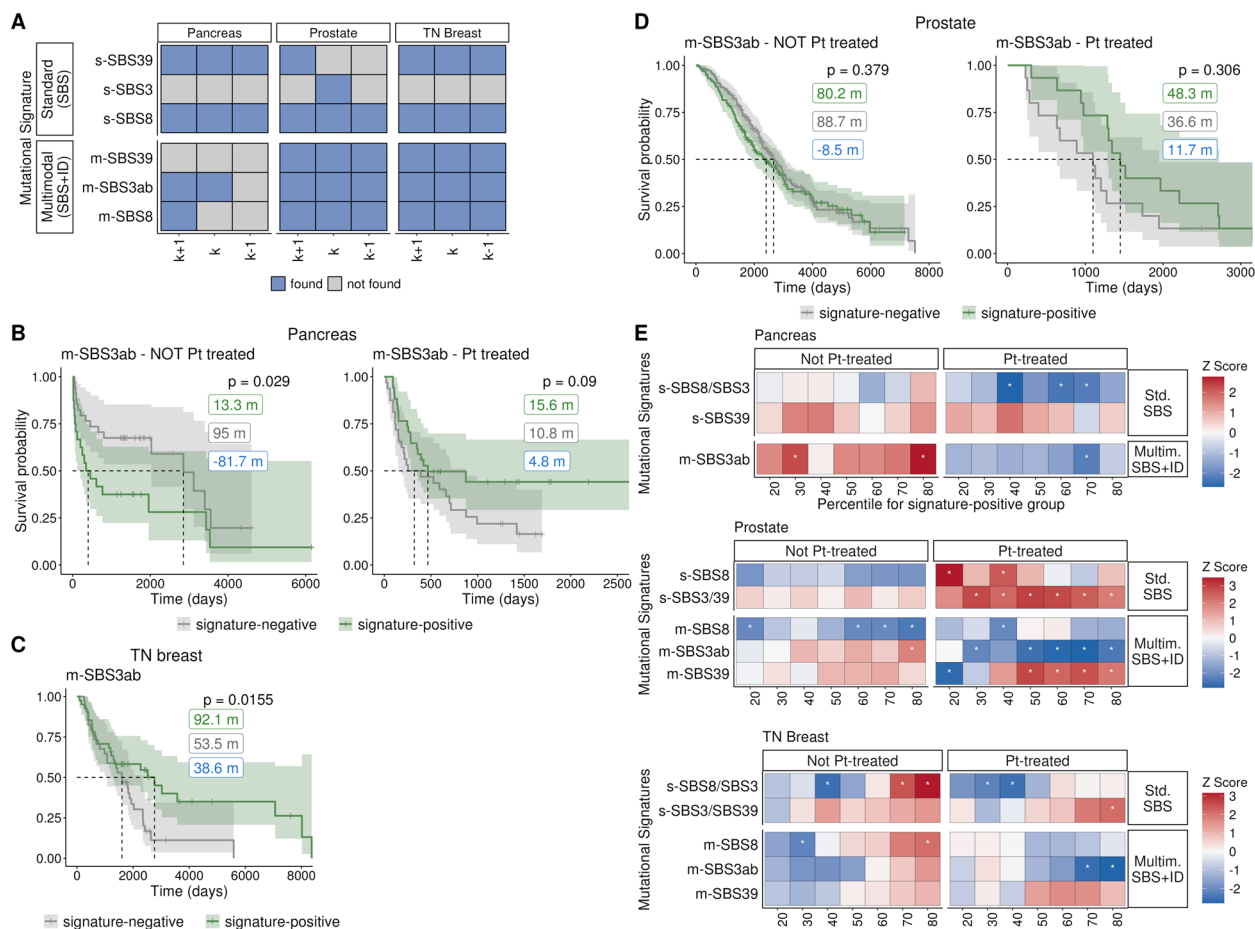


Fig. 5 Multimodal SBS3 signatures in prostate, pancreas and breast cancer. **A** The multimodal method allows consistent extraction of SBS3, SBS8, and SBS39 in various tissue types. “Found” implies that a de novo extraction with NMF resulted in one of the extracted spectra being individually assigned to a certain signature. **B** Kaplan–Meier curves for signature m-SBS3ab for pancreatic cancer Hartwig samples, separated into platinum treated and platinum-untreated. The median survival (in months) is shown for both groups, as well as the difference between them in blue. **C** Kaplan–Meier curves for signature m-SBS3ab for TN breast cancer samples from Hartwig. **D** Kaplan–Meier curves for prostate cancer Hartwig samples, separated into platinum-treated and platinum-untreated. **E** Results of multivariate Cox regression for standard and multimodal signatures, separated into platinum-treated and platinum-untreated patients of TN breast, prostate, and pancreatic cancer. The x axis shows the different percentiles tested to classify samples by signature activity, and the asterisks represent a significant link between the variable and overall survival ($p < 0.05$)

for m-SBS3ab than for s-SBS8/SBS3 (Additional File 1: Fig. S21).

Given prior reports of improved response to platinum therapy in HR deficient cancers, e.g., for pancreas [62, 63], we further hypothesized that the SBS3 signatures will predict survival specifically in platinum-treated patients of different cancer types. In this data set, prostate and pancreas cancer patients treated with platinum have much lower survival probability, likely reflecting patient selection effects such as platinum being applied to more advanced or aggressive tumors. To account for stratification reflected in the treatment received by the patients, we divided the cohort into signature-positive, signature-negative, and platinum-treated and platinum-untreated,

comparing their median survival time (Fig. 4b, d). For prostate cancer, signature-positive patients treated with platinum showed a higher median survival time. This effect was stronger for m-SBS3ab signature, with a difference in median survival of 11.7 months (not significant), than for s-SBS3 signature, where the difference was less than a month (not significant) (Additional File 1: Fig. S21). This contrasts with the survival results for the platinum-untreated patients, where signature-positive patients for both s-SBS3 and m-SBS3ab had a much lower median survival time (not significant).

Finally, for pancreatic cancer, we also see that in the platinum-untreated group, SBS3 signature-positive patients actually show lower median survival times,

while this trend is reversed for the platinum-treated group where both s-SBS3 (Additional File 1: Fig. S21) and m-SBS3ab (Fig. 5d) predict longer survival. M-SBS3ab signature-positive patients treated with platinum show a median survival of 4.8 months more than the signature-negative group, while in the platinum-untreated patients there is around a 82-month difference in the other direction (Fig. 5d).

To account for the impact of other mutational signatures and age, we performed a multivariate Cox regression for these tissues (Fig. 5e), separately analyzing platinum-treated and platinum-untreated patients. For TN breast cancer, m-SBS3ab signature is associated with better overall survival, with a minimum Z-score of -1.89 for platinum-untreated patients ($p = 0.09$) and -2.78 for platinum-treated patients ($p = 0.005$). For prostate and pancreas cancer, we again observe the previously described trend: m-SBS3ab-positive status predicts an increased survival probability in platinum-treated patients, but not in the platinum-untreated patients. For example, the minimum Z-score for m-SBS3ab in prostate cancer decreases from -0.25 ($p = 0.80$) to -2.84 ($p = 0.004$) when only studying platinum-treated patients (Fig. 5e). Similarly, for pancreas cancer the minimum Z-score changes from 0.12 ($p = 0.90$) to -2.19 ($p = 0.028$) (Fig. 5e). This improvement in survival probability is not observed for the other of the signatures studied, SBS8 and SBS39 (Fig. 5e), indicating that this effect is specific to SBS3 (also seen in the mixed signature s-SBS8/SBS3) and that we can capture the positive effect of platinum treatment in HR deficient patients using SBS3 as a marker. In fact, both standard and multimodal SBS39 are significantly associated with shorter survival in the platinum-treated group (Fig. 5e), further supporting that SBS39 (probably equivalent to SBS3c, see above) has origins that do not involve HRd.

Adjusting for other mutational signatures and age, included in the regression, allowed us to uncover the positive association of m-SBS3ab with survival in several instances. For example, in pancreatic cancer, older patients (over 65) have a better increase in survival probability linked with m-SBS3ab, than younger patients do (Additional File 1: Fig. S22b), with a difference in median survival of 9.05 months higher for the m-SBS3ab positive. However, in the younger group, the results are non-significant and the effect size is opposite (28 months less). For prostate, platinum-treated m-SBS3ab-positive older patients also showed a stronger increase in survival linked with m-SBS3ab than the younger patients. This was true for some of the thresholds tested, for example 70 years old and 50 % percentile for signature activity (Additional File 1: Fig. S22a). However, these results are more modest than in pancreas cancer and less consistent

throughout other thresholds (Additional File 1: Fig. S22a).

Similarly to what was done for age, we stratified patients by the presence of SBS3 and another signature of interest. We found examples where signature m-SBS36 in pancreas, and m-SBS13 in prostate (Additional File 1: Fig. S23) have statistical interactions with m-SBS3ab in predicting survival. We emphasize that these results should be interpreted with care, as the sample sizes of some of the subgroups created for the Kaplan–Meier curves are small. However, this is reflected in the confidence interval (CI) bands of the curves and the p -values.

Overall, the multimodal approach generated prognostic marker SBS + ID signatures applicable across three additional tumor types commonly displaying HRd. Importantly, the link between SBS3 and a better prognosis became clearer when looking only at patients treated with platinum drugs, and there is suggestive data that age and the other mutational signatures modulate the association of m-SBS3ab with survival.

Replication timing-aware extraction clarifies etiology of multimodal mutational signatures

Mutational processes act differently in different domains of the genome, in association with replication timing (RT). We adapted an approach to refine mutational signatures by incorporating information on early and late RT domains into the signature inference process (Fig. 6a). This will help clarify the real etiology of the signatures, since HRd signatures were enriched in early-replicating RT domains [64, 65], as defined from Repli-Seq [53] data (Methods). This method was applied to three cohorts of ovarian cancer: DECIDER, Hartwig, and PCAWG; please see Additional File 3: Text S2 and Additional File 1: Fig. S24 and 25 for detailed results. In brief, the 192 channel RT-aware signatures (each of the 96 SBS split into early and late RT bins) did improve extraction of unimodal SBS3. However, RT information did not appreciably improve extraction of the multimodal, indel-aware m-SBS3b signature. We suggest that adding RT information to mutational signature inference may be helpful in the cases when indels are unreliable (e.g. whole-exome sequencing), or with smaller cohorts of WGS (comparison of individual cohorts is addressed in the Additional File 3: Text S3).

Multimodal MMRd signatures are prognostic in colorectal cancer

We extracted standard SBS signatures, ID signatures, and multimodal SBS + ID signatures from 1800 colorectal cancer samples, from Hartwig cohort [39], together with the Uppsala University large-scale WGS [44] Swedish cohorts.

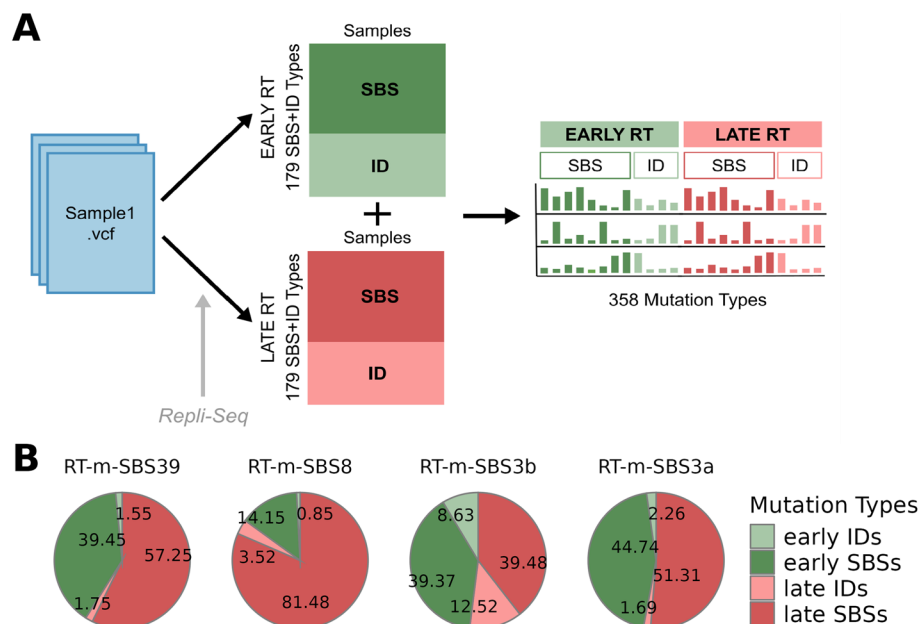


Fig. 6 Stratifying by DNA RT to improve mutational signature extraction. **A** Schematic showing how RT-multimodal signatures are derived by dividing mutations in each tumor sample into two groups depending on their location within late or early RT regions in the genome, generating 358 mutation types of multimodal RT-aware signatures. **B** SBS and ID composition in early and late RT within the RT-multimodal mutational signatures of interest

We focused on the analysis of various mutational signatures associated with microsatellite instability (MSI), a type of genome instability that indicates that DNA mismatch repair is deficient. MSI and/or MMRd have been associated with lower cancer aggressiveness and better response to immunotherapy, while possibly worse responses to certain chemotherapies (reviewed in [66]). Therefore, we asked if the multimodal signature extraction method could improve MMR signatures in their ability to predict patient outcomes in these two large cohorts of colorectal cancers. We extracted 5 unimodal SBS-only signatures with putative links with MMRd (based on containing one of the SBS previously reported as MMR-associated: SBS6, 15, 21, 26, and 44) [8, 67], and 7 multimodal SBS + ID signatures putatively linked with MMRd. Moreover, we extracted 5 ID-only MMRd-associated signatures (containing ID1, ID2, and ID7) [1] (see signature spectra in Additional File 2: Table S1-S3).

We performed Cox regression analysis for the MMRd signatures, including all MMRd-associated SBS signatures and SBS + ID signatures in a joint regression, further adjusting for patient age, stage, and cohort (Hartwig *versus* Uppsala) (Fig. 7a). Herein, we observed how only the multimodal signature m-SBS44 + ID7 was significantly associated with better survival at a threshold $p < 0.05$ (HR = 0.8, $p = 0.022$ and HR = 0.81, $p = 0.019$) (see signature exposures in Additional File 2: Table S14). However, this significant survival association was not

observed for the two standard s-SBS44-like signatures (Fig. 7a). Interestingly, in this multivariable analysis, we observed a significant association with worse survival of m-SBS44 + 57 + ID2 throughout several of the thresholds tested. Both ID2 and ID7 are reported MMRd-associated spectra [1], where ID2 is exclusively composed of 1nt deletions at homopolymer T runs, and ID7 is a mixed spectrum of 1nt deletions enriched at poly-C over poly-T, as well as 2nt deletions at homopolymer stretches. This analysis highlights the role of the ID spectra in the multimodal mutational signature extraction and its relevance for prognostic prediction. However, regarding ID signatures, only s-ID2 showed a significant association with worse survival, with a maximum HR of 1.41 ($p = 0.001$) (Additional File 1: Fig. S26). In contrast, signature s-ID7 did not associate with longer survival time, as m-SBS44 + ID7 did. Overall, these results suggest that, in the context of MMRd, multimodal signatures offer an advantage for survival analysis compared to SBS-only or ID-only signatures. As for the rest of MMRd related signatures, we do observe consistent associations only for signatures s-SBS21 and m-SBS21 (Fig. 7a). They show significance in one and two of the binarization thresholds (into signature-positive vs signature-negative) tested, respectively, being associated with longer survival. Next, we studied association of mutational signatures with immunotherapy responder patients (here defined as a partial response (PR) or stable disease (SD)) versus non-responders

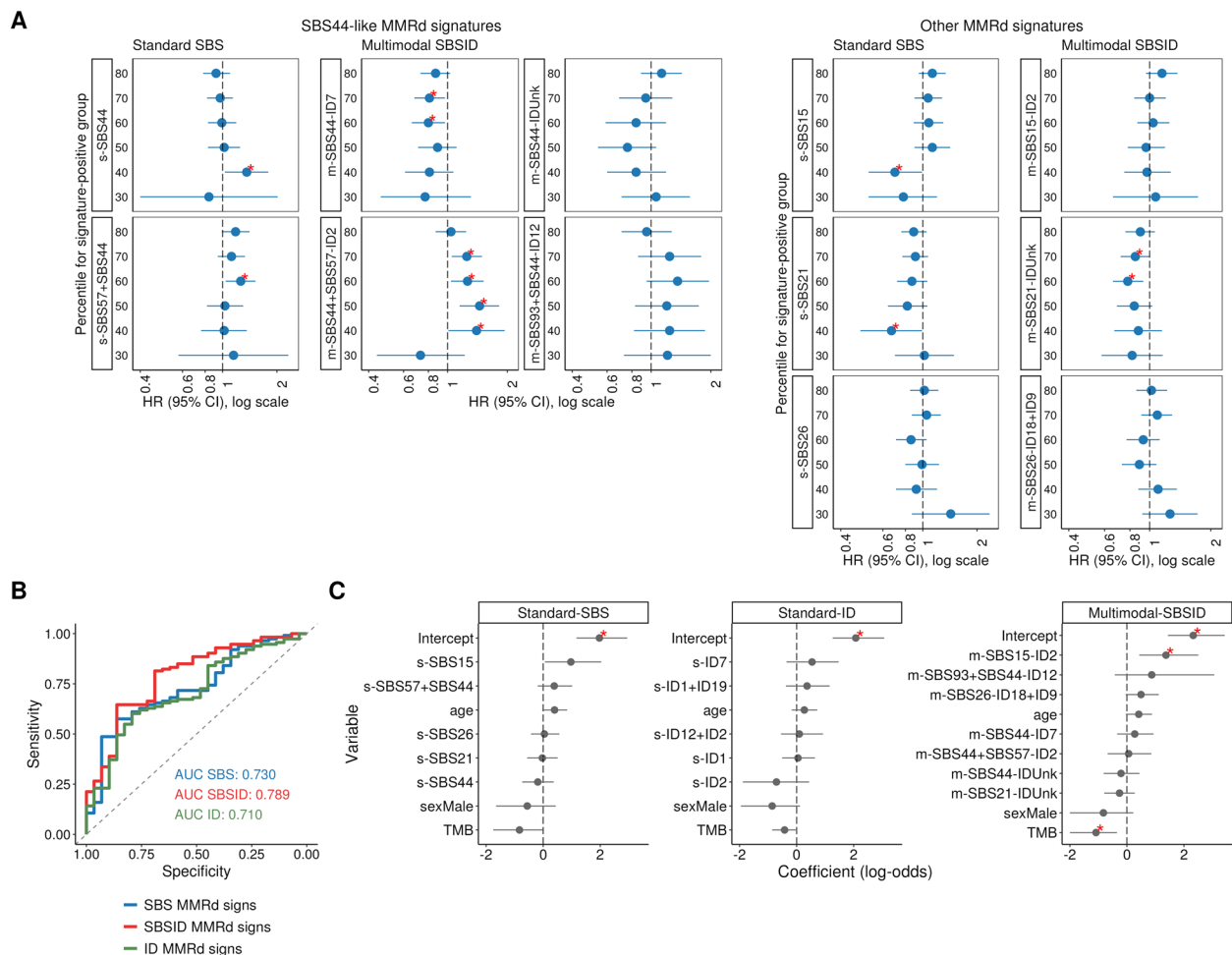


Fig. 7 Multimodal signatures of MMRd predict survival and immunotherapy response in colorectal cancer. **A** Results of multivariate Cox regression for standard and multimodal MMRd-related signatures (SBS44, SBS21, SBS15, and SBS26) in colorectal cancer. The MMRd-related signatures, as well as age, stage, and cohort were included in the Cox regression. The plot shows the hazard ratio (HR, with 95% CI) and significance, where asterisks imply $p < 0.05$ for different thresholds (30th to 80th percentile) used to classify samples into signature-positive or signature-negative. "Unk" in the signature names refers to a mutational signature that SigProfiler could not map to any of the known COSMIC signatures. **B** ROC curves from logistic regression predicting immunotherapy response (partial response or stable disease vs. progressive disease) using multimodal and standard SBS and ID-only mutational signatures. The models included MMRd-related mutational signatures, tumor mutation burden (TMB), age, and sex. The plot shows the AUCs for each model. **C** Model coefficients from logistic regression predicting immunotherapy response using multimodal and standard SBS and ID-only mutational signatures. The plot shows the coefficients, with 95% CI and significance, where asterisks imply $p < 0.05$.

(progressive disease (PD)) in the Hartwig cohort, restricting to the patients that received immunotherapy. We used the MMRd-related signature activities (scaled) to predict response to immunotherapy in a logistic regression, also including TMB, age, and sex variables to the model. The area under the ROC curve (AUC) for the multimodal signatures is 0.789, while for the standard SBS-only signatures it is 0.730, and for ID-only signatures it is 0.710 (Fig. 7b). The model coefficients show m-SBS15 as a significant variable ($p = 0.009$) for responders, exceeding s-SBS15 (Fig. 7c). We note m-SBS26 is trending in the same direction ($p = 0.093$). The overall

good survival signature m-SBS44 does not predict significantly better immunotherapy responses in specific. Overall, multimodal mutational signatures are able to better predict survival and immunotherapy response in colon cancer patients based on MMRd. This demonstrates applicability of the method to other cancer types and other mutational signatures

Discussion

In this study, we developed and tested a multimodal approach to mutational signature analysis that jointly considers SBS and indels. Applying this method to WGS

data from HGSOC, we found that the HRd-associated signature SBS3 can be robustly separated into two sub-signatures: m-SBS3a linked to *BRCA1* deficiency and containing the ID8 indel spectrum, and m-SBS3b linked to both *BRCA1* and *BRCA2* deficiency and containing the ID6 indel spectrum, composed of deletions flanked by 2–4-nt-long microhomology. Importantly, m-SBS3b was a strong predictor of improved overall survival in HGSOC patients across four independent cohorts. The clear differential association of these two signatures with *BRCA1* and *BRCA2* variants, as well as their different predictive value for survival, indicate that they are not products of “oversplitting” the combined m-SBS3ab signature, but rather that they reflect real biological differences. The third signature we identified, m-SBS3c, is probably the same signature detected previously as SBS39. This likely does not represent HRd as such, but might still reflect a related process, for instance the increased TLS usage which can occur downstream of the HRd [29].

While SBS3 has been previously associated with HRd in various studies [1, 26, 27], the “flat” profile of this signature makes it challenging to distinguish it from SBS8 and from other widespread mutational processes like SBS5 and SBS40 [2]. By incorporating indel information, specifically the microhomology-mediated deletions characteristic of HRd captured in ID6 and ID8 [26], we more precisely inferred exposures to SBS3 and identified HRd tumors. We analysed whether a similar multimodal strategy could refine other mutational signatures with indel components, for instance signatures of MMR deficiency, which also carry therapeutic implications (reviewed by Mas-Ponte et al., 2022 [66]). When indels are not available or accurate (for instance in whole-exome sequencing), inclusion of DNA replication time during mutational signature inference can improve its accuracy, based on differential enrichment of mutational processes across chromosomal domains [68].

Adding other mutation types beyond SBS and indels, like CNA or more generally various types of DNA rearrangements (structural variants, SVs), could further enhance signature detection for HRd and for other clinically or biologically interesting mutational processes. Rearrangement features already underpin the specialized algorithms HRDetect [34] and CHORD [23] for HRd detection by supervised machine learning. Further studies reported the use of rearrangement features in mutational signatures, for instance where SNVs were considered jointly with structural variants. They suggested how a joint analysis can make signature inference more reliable with sparse SV data, demonstrating survival prediction in one cohort of HGSOC [56]. Another

study analyzed multidimensional mutational signatures, where SNVs were decomposed by various types of genomic regions (including replication time, and chromatin states). This “tensor signatures” approach identified a SBS3-like SNV signature (“TS19”), with an accessory spectrum of structural variants, in particular tandem duplications and deletions >10 nt, enriched in ovarian and breast cancers [69]. Moreover, the link of copy number signatures to different DNA repair failures was reported [70, 71]; further work needs to study if this information contained within rearrangement/CNA signatures add information beyond SBS + ID profiles, or merely recapitulate them in signature extraction and to deriving prognostic predictions.

As regards the strengths of the study, they include: (i) methodological advance for improving mutational signatures via an easy-to-implement modification, which can be retrofitted to existing tools such as SigProfiler or MuSiCal, both for de novo extraction and for signature refitting; (ii) evidence of strong prognostic impact of HRd signatures refined via multimodal method, replicating in four independent HGSOC cohorts, besting existing genetic markers; (iii) identification of subtypes of the SBS3 signature, which is decomposed into SBS3a and SBS3b having distinct mechanistic causes and prognostic relevance; (iv) the ability of our approach to extract more stable signatures from smaller sample sets is also advantageous, enabling characterization of rarer tumor types or subtypes to discover new markers based on mutational signatures; (v) we provide a proof-of-concept for application of the multimodal signatures to other cancer types in addition to HGSOC, and to classifying other mutagenic processes (MMRd), aligning with the growing evidence for the relevance of various DNA repair deficiency signatures to predicting therapy response in a variety of settings [12, 16, 57, 72–74].

Our study also has some limitations, which highlight directions for future work: (i) the analyses focus on WGS data; however, various clinical datasets may use panel sequencing or exome sequencing, where indel mutations are sparse and the multimodal SBS + ID approach would likely not apply; (ii) while associations of m-SBS3b with better prognosis replicate across various cohorts, another step towards clinical application is to systematically vary different technical parameters including quality and purity of material, DNA sequencing technology and depth, and bioinformatics pipelines for variant calling to estimate robustness of markers; (iii) it remains to be elucidated what is the mechanistic basis for the stronger prognostic value of m-SBS3b compared to m-SBS3a; speculatively, the *BRCA2* loss-of-function might trigger additional DNA replication or repair problems in

addition to HRd, rendering tumors more vulnerable to therapy; however, this needs to be followed up with functional studies; (iv) the tentative associations of other mutational signatures with prognostic outcomes (including platinum SBS31 and the unknown etiology SBSUnk-ID9 in HGSOc, as well as SBS36 and SBS13 in other tumor types) suggest existence of modifiers of HRd-related outcomes, which will need larger and/or better-annotated datasets to be studied in depth. In conclusion, our work demonstrates the power of a simple multimodal mutational signature approach to more accurately measure clinically relevant mutational processes in cancer due to DNA repair deficiencies.

Conclusions

Our study demonstrates that incorporating both SBS and ID features into mutational signature analysis improves the resolution and clinical relevance of HRd-associated signatures. By decomposing SBS3 into distinct sub-signatures with different mechanistic origins and prognostic value, we reveal biologically meaningful heterogeneity within HRd. Our approach outperforms known HRd markers in predicting survival and is easily adaptable to existing tools, making it accessible for broader application. These findings highlight the potential of multimodal mutational signatures as robust biomarkers for therapy stratification in cancer.

Abbreviations

AUC	Area under the ROC curve
CI	Confidence interval
CNA	Copy number alteration
GIS	Genomic instability score
HGSOc	High-grade serous ovarian carcinoma
HR	Hazard ratio
HR	Homologous recombination
HRd	Homologous recombination deficiency
ID	Indel
<i>k</i>	Number of extracted mutational signatures
LOH	Loss of heterozygosity
LST	Large-scale state transitions
MMRd	Mismatch repair deficiency
MSI	Microsatellite instability
NER	Nucleotide excision repair
NMF	Non-negative matrix factorization
NNLS	Naïve non-negative least squares
PARPi	Poly(ADP-ribose) polymerase inhibitor
PCA	Principal component analysis
PD	Progressive disease
PFS	Progression-free survival
PR	Partial response
RT	Replication timing
SBS	Single base substitution
SD	Stable disease
SNV	Single nucleotide variant
SV	Structural variant
TAI	Telomeric allelic imbalances
TMB	Tumor mutation burden
TN	Triple-negative
VIF	Variance inflation factor

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13073-025-01497-7>.

Additional File 1. PDF file (.pdf): Supplementary Figures S1–S26: Figure S1. Signature spectrum for SBS + ID multimodal signatures and COSMIC standard signatures. Figure S2. Signature activity correlations. Figure S3. Association of ID signatures with HR deficiency. Figure S4. Associations of genomic scores with genetic variants in major HR genes. Figure S5. Spearman correlation between HR deficiency related signature activity and number of duplications. Figure S6. Spearman correlation between HR deficiency related signature activity and number of SV deletions. Figure S7. Survival Kaplan–Meier curves for HR deficiency related signatures. Figure S8. Survival Kaplan–Meier curves for other related signatures. Figure S9. Forest plots from Cox multivariate regression show relevance of HR deficiency signatures to survival. Figure S10. Forest plots from Cox multivariate regression showing relevance of variables to survival. Figure S11. Survival Kaplan–Meier curves for age and stage in ovarian cancer. Figure S12. Overall survival and progression-free survival prediction comparison. Figure S13. Complementarity between HR deficiency mutational signatures in predicting ovarian cancer survival. Figure S14. Analysis of standard and multimodal signatures of HR deficiency in four cohorts of ovarian cancer. Figure S15. Mutational signature m-SBS3ab obtained by MuSiCal refitting method. Figure S16. Association of CNA states to survival in ovarian cancer. Figure S17. Mutational signatures m-SBS3a and m-SBS3b obtained by assignment (refitting) method with SigProfiler. Figure S18. HR deficiency and survival analysis of multimodal signature m-SBS3ab obtained by MuSiCal refitting method. Figure S19. Survival associations of genomic scores and multimodal mutational signature m-SBS3b. Figure S20. Association of standard and multimodal signatures to HR deficiency in other tissues. Figure S21. Survival Kaplan–Meier curves for s-SBS3 in other tissues. Figure S22. Interactions between m-SBS3ab and age in predicting survival. Figure S23. Interactions between m-SBS3ab and other mutational signatures in predicting survival. Figure S24. RT-aware multimodal signature method allows the extraction of m-SBS3a and m-SBS3b. Figure S25. RT-aware signatures are strong predictors of survival. Figure S26. Results of multivariate Cox regression for standard ID MMRd-related signatures in colorectal cancer.

Additional File 2. multi-sheet Excel file (.xls): Supplementary Tables S1–S14: Table S1: Mutational signature spectrum of relevant SBS standard mutational signatures extracted from the ovarian and other cancer types (breast, pancreas, prostate and colorectal) cohorts. Table S2. Mutational signature spectrum of relevant ID standard mutational signatures extracted from the ovarian and colorectal cohorts. Table S3. Mutational signature spectrum of relevant SBS + ID multimodal mutational signatures extracted from the ovarian and other cancer types (breast, pancreas, prostate and colorectal) cohorts. Table S4. Exposures of the relevant mutational signatures for the pan-HGSOc analysis samples. Table S5. Results for multivariate Cox regression in the pan-HGSOc cohort for relevant standard SBS, ID and multimodal SBS + ID signatures. Table S6. Exposures of the relevant mutational signatures for the DECIDER HGSOc samples. Table S7. Exposures of the relevant mutational signatures for the PCAWG HGSOc samples. Table S8. Exposures of the relevant mutational signatures for the Hartwig HGSOc samples. Table S9. Exposures of the relevant mutational signatures for the OVCARE HGSOc samples. Table S10. Results for multivariate Cox regression comparing multimodal signature m-SBS3b and other HR detection methods scores/probabilities. Table S11. Exposures of the relevant mutational signatures for the TN breast Hartwig samples. Table S12. Exposures of the relevant mutational signatures for the prostate Hartwig samples. Table S13. Exposures of the relevant mutational signatures for the pancreas Hartwig samples. Table S14. Exposures of the relevant mutational signatures for the colorectal Uppsala University and Hartwig samples.

Additional File 3, PDF file (.pdf): Supplementary Text S1–S3: Text S1: Extraction and etiology of m-SBS3c and m-SBS39. Text S2: Association of mutational signatures to HR deficiency in TN breast, pancreas and prostate cancer. Text S3: Analysis of RT-aware mutational signatures.

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Author contributions

P.F. collected and curated the data, performed most analyses, visualized the data, and interpreted the results. I.G. contributed to the conceptual development, supervised parts of the analysis, performed data collection, and mutation annotation. F.S. conceptualized the analysis, devised the overall methodology, and supervised the study. F.S. and P.F. drafted the manuscript jointly. All authors read and approved the final manuscript.

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Data Availability

In this study, published tumor datasets were analyzed. We obtained the WGS somatic mutation calls for the Hartwig Medical Foundation study [<https://www.hartwigmedicalfoundation.nl/en/>] under request DR-260 (Hartwig data is available under restricted access via <https://www.hartwigmedicalfoundation.nl/en/data/data-access-request/>) [75], the PCAWG study re-processed by the HMF pipeline at the International Cancer Genome Consortium (ICGC) data portal [<https://docs.icgc-argo.org/docs/data-access/icgc-25k-data>] [76], the DECIDER data from within a collaborative project [<https://www.deciderproject.eu/>] also available under restricted access via EGA [<https://ega-archive.org/studies/EGAS00001006775>] [77], the OVCARE data from the British Columbia Ovarian Cancer Research Program [<https://ega-archive.org/studies/EGAS00001002390>] [78], and the Uppsala University colorectal short somatic variant call data obtained via European Variation Archive under accession number PRJEB61514 [<https://www.ebi.ac.uk/ena/browser/view/PRJEB61514>] [79]. CHORD probabilities were downloaded from Nguyen et al. study [<https://doi.org/10.1038/s41467-020-19406-4>] [23], HRDetect probabilities from Davies et al. study [<https://doi.org/10.1038/nm.4292>] [34], GLS scores from Telli et al. study [<https://doi.org/10.1158/1078-0432.CCR-15-2477>] [48], and CN Signature 3 exposures from Macintyre et al. study [<https://doi.org/10.1038/s41588-018-0179-8>] [43]. Relevant data generated during this study are included in this published article as Additional File 2. Code necessary to extract multi-modal (SBS+ID) mutational signatures is available at https://github.com/patriciaferrer/multimodal_mut_signs_SBSID [80].

Availability of data and materials

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Institute for Research in Biomedicine (IRB Barcelona), The Barcelona Institute for Science and Technology (BIST), Carrer Baldori Reixac 10, 08028 Barcelona, Spain. ²Biotech Research and Innovation Centre (BRIC), Faculty of Health and Medical Sciences, University of Copenhagen, Ole Maaløes Vej 5, 2200 Copenhagen, Denmark. ³Catalan Institution for Research and Advanced Studies (ICREA), Passeig de Lluís Companys, 23, 08010 Barcelona, Spain.

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